BITS PILANI | GOA CAMPUS

PROJECT ANTARA

Mission Concept Review

3U CubeSat for In-Situ Characterization of the
South Atlantic Anomaly Radiation Environment

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1 Mission Context, Objectives, and Success Criteria

1.1 Executive Summary

Project ANTARA is a 3U CubeSat mission at BITS Pilani, K.K. Birla Goa Campus, dedicated to the in-situ characterization of the South Atlantic Anomaly (SAA) radiation environment. Operating in a 600 km Sun-Synchronous Orbit (97.6° inclination), the spacecraft carries a custom Flux Particle Detector Unit (FPDU) payload to measure unidirectional differential proton flux across the 5–100 MeV range in a proposed multi-channel energy reconstruction.

The mission is structured around three tiers of success that scale spatial resolution, coverage, and scientific ambition. Beyond its scientific objective, ANTARA serves as a student satellite training platform, exposing undergraduate students to the complete spacecraft development lifecycle from concept through operations.

This document represents the Mission Concept Review (MCR) package for Project ANTARA, consolidating mission requirements, architecture, orbit design, operational concept, subsystem design status, and simulation results.

1.2 Mission Need

The ANTARA mission has been conceptualized to address a combination of scientific, educational, and national-level needs associated with the growing importance of space-based radiation studies and indigenous small-satellite development. Modern spacecraft operating in Low Earth Orbit (LEO) are increasingly exposed to harsh radiation environments that can significantly affect mission reliability, onboard electronics, detector systems, and long-term spacecraft survivability. Among these environments, the South Atlantic Anomaly (SAA) represents one of the most severe and scientifically significant radiation regions encountered by satellites in LEO.

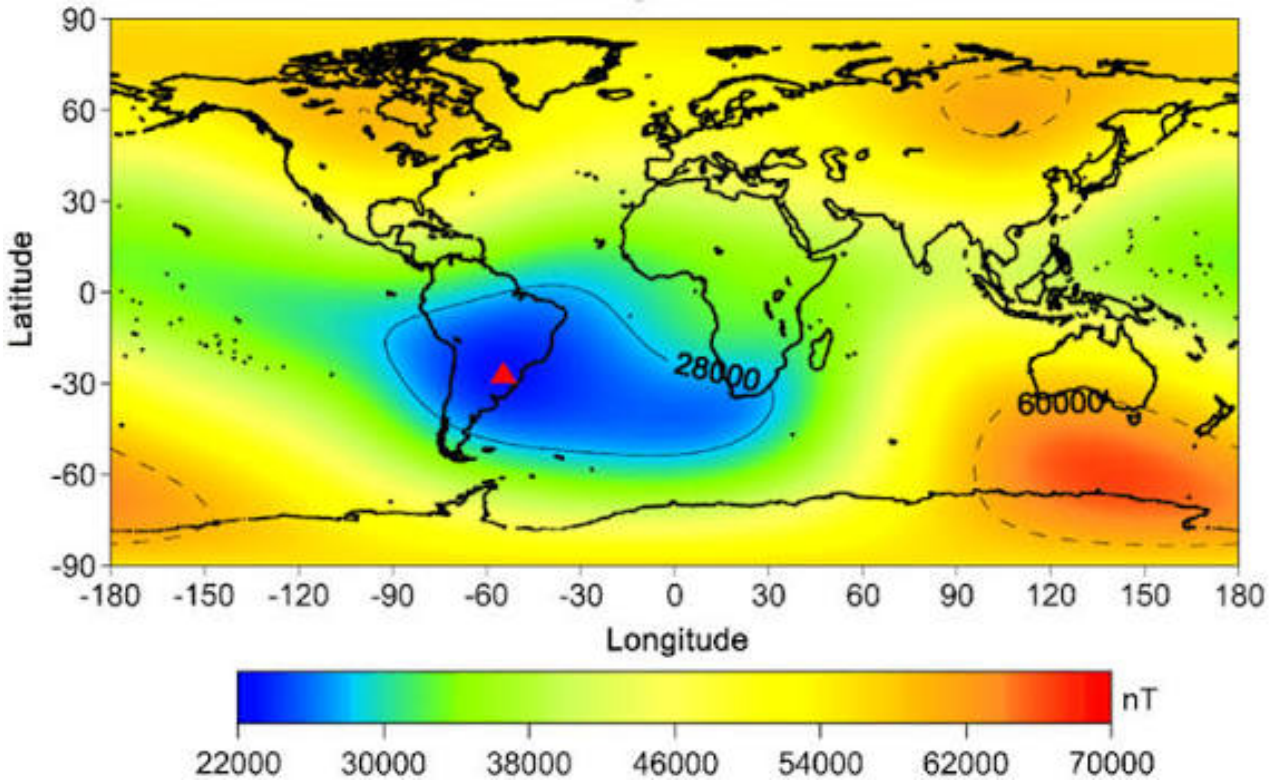


Figure 1: South Atlantic Magnetic Anomaly (year 2000; source image from original draft).

Although several trapped-radiation models such as AP8 and AP9 currently exist, substantial uncertainties remain regarding temporal proton-flux variability, storm-time behavior, directional anisotropies, and localized radiation structures within the SAA. Existing models are primarily statistical and often lack sufficient real-time data required for modern small-satellite missions and future radiation-aware spacecraft operations.

The ANTARA mission seeks to contribute toward closing these gaps through direct in-situ measurement of high-energy proton populations using a dedicated CubeSat operating in a Sun-Synchronous Orbit. The project also enables students to gain practical experience in CubeSat designing and modelling.

1.2.1 Educational Motive

One of the primary motivations behind ANTARA is the creation of an educational and research platform capable of providing hands-on experience in real-world spacecraft development.

The mission enables participating students to experience all major phases of aerospace project development. Unlike isolated academic laboratory experiments, CubeSat missions introduce students to highly constrained engineering environments where mass, power, thermal limits and others interact simultaneously. This systems-level complexity is one of the most important educational benefits of the ANTARA mission.

1.2.2 Scientific Motive

The primary scientific motivation behind ANTARA is the need for improved understanding of the trapped proton environment within the South Atlantic Anomaly and its implications for spacecraft operations, radiation modelling, and near-Earth space science.

The South Atlantic Anomaly represents the region where the inner Van Allen radiation belt approaches closest to Earth due to the offset and tilt of Earth's magnetic dipole. Spacecraft traversing this region are exposed to significantly elevated fluxes of energetic charged particles, particularly high-energy protons capable of producing Single Event Upsets (SEUs), sensor noise, detector saturation, memory corruption, semiconductor degradation, and cumulative radiation damage.

ANTARA aims to contribute toward this scientific need by directly measuring Unidirectional Differential Proton Flux (FPDU) within the energy range of 5 MeV to 100 MeV using an onboard proton detection payload. The mission seeks to generate spatially and temporally correlated radiation datasets that can be compared against existing radiation models including AP8 and AP9.

1.2.3 National Motive

From a national perspective, ANTARA aligns strongly with India's growing emphasis on indigenous space technology development, scientific self-reliance, and advanced research capability. The mission contributes toward India's requirement for indigenous expertise in spacecraft systems engineering, radiation-aware electronics, mission operations, and advanced satellite technologies.

Also, the satellites in the past few years have faced various anomalies with this region due to which communication signals with them were lost. Data collected by ANTARA maps the shifting borders of the SAA, allowing Indian satellites to shield or power down domestic defense and communication satellites before they sustain damage. ANTARA provides an opportunity

for low-cost effective and indigenous space tech to be tested via its CubeSat.

1.3 Mission Significance

1.3.1 Primary Flight Objective

The primary objective is to measure and map unidirectional differential proton flux in the South Atlantic Anomaly over the 5–100 MeV range.

1.3.2 Secondary Mission Objectives

The secondary objectives are to examine East–West asymmetry, characterize the bifurcation of the SAA, investigate atmospheric-loss coupling, and develop corrections to radiation-model predictions from ANTARA measurements. Storm-time comparisons remain an exploratory objective dependent on available observations. The detector architecture and channel definition remain under development; the science-success criteria are proposed targets to be tested against actual detection and sampling capability.

1.4 Mission Success Criteria

Mission success criteria define the measurable conditions under which the ANTARA mission can be considered operationally and scientifically successful. These criteria establish a structured framework for evaluating mission performance at different levels of achievement throughout the mission lifecycle. The criteria are divided into three progressive categories:

- Minimum Mission Success (MMSC)
- Full Mission Success (MSC)
- Extended Mission Success (EMSC)

1.4.1 Minimum Mission Success Criteria (MMSC)

The MMSC defines the lowest acceptable bar for declaring the mission a scientific and engineering success. Meeting the MMSC demonstrates that ANTARA has survived launch, commissioned its payload, operated within the SAA environment, and produced scientifically interpretable, calibrated flux data at coarse spatial resolution.

Table 1: MMSC mission duration targets

Parameter	Target
Mission duration	4-6 Months minimum
Orbital count	1750 orbits (4 months) to 2650 orbits (6 months)
SAA transit count	700 to 1050 SAA transits across the mission duration
Minimum SAA passes for MMSC science	$\geq$ 150 passes with payload operational and telemetry received
Definition of 'operational'	Coincidence system functional, $\geq$ 2 energy channels recording events, housekeeping telemetry nominal

Table 2: MMSC spatial mapping requirements

Parameter	Value
Grid resolution	5° × 5° in geographic latitude and longitude
SAA bounding box	270°W to 10°E, 50°S to 10°N (130° × 60°)
Total grid cells in bounding box	26 × 12 = 312 cells
Cell dwell time per pass	~74 seconds per 5° latitude band at 7.5 km/s

Table 3: MMSC science requirements

ID	Requirement
MM-M-01 REQUIRED	The mission shall produce a calibrated 5°×5° flux map covering at least 70% of grid cells within the SAA bounding box with valid measurements (≥3 independent SAA passes per cell).
MM-M-02 REQUIRED	Coverage shall include the SAA flux core (centred approximately on 310°E, 30°S) to ≥90% even if peripheral cells fall below the 70% global threshold.
MM-M-03 REQUIRED	Each populated spatial cell shall report a calibrated differential proton flux value in units of particles/cm²/s/sr/MeV for each operational energy channel.
MM-M-04 REQUIRED	σ_F/F ≤ 50% per cell per channel. Requires approximately N ≥ 4 counts per cell per channel.
MM-M-05 REQUIRED	Flux maps shall be produced for a minimum of 2 energy channels independently.

MMSC Scientific Deliverables

- Calibrated 5°×5° differential proton flux map in at least 2 energy channels
- Flux uncertainty map (σ_F/F per cell per channel)
- Pass-count coverage map (N_passes per spatial cell)
- Detector performance report: channel stability, livetime fractions, dead-time characterisation
- model-baseline comparison plots with residual map at 5° resolution
- Mission timeline with all anomaly flags and excluded data intervals documented

1.4.2 Mission Success Criteria (MSC)

Table 4: MSC mission duration targets

Parameter	Value
Required duration	12 months - 14 months minimum
Orbital count	5400 orbits (12 months) to 6300 orbits (14 months)
SAA transit count	2150 to 2500 SAA transits
Minimum SAA passes for MSC science	≥ 900 passes with payload operational

Table 5: MSC spatial mapping requirements

Parameter	Value
Grid resolution	1° × 1° in geographic latitude and longitude
Total grid cells in bounding box	130 × 60 = 7,800 cells
Cell dwell time per pass	15 seconds per 1° latitude band at 7.5 km/s

Table 6: MSC science requirements

ID	Requirement
MS-M-01 REQUIRED	The mission shall produce a calibrated 1°×1° flux map covering at least 90% of SAA bounding box cells that contain detectable flux.
MS-M-02 REQUIRED	Each populated 1°×1° cell shall achieve σ_F/F $\leq$ 25% in at least the two lowest-energy operational channels. Requires N $\geq$ 16 counts per cell per channel.
MS-M-03 REQUIRED	Independent flux maps for all operational energy channels, co-registered on the same 1°×1° grid, released as a unified multi-band atlas.
MS-M-04 REQUIRED	Minimum of 5 independent SAA passes per populated cell for inclusion in primary science map.
MS-M-05 REQUIRED	Monthly-averaged sub-maps produced and compared. Month-to-month variation exceeding 3σ in any cell shall be flagged.

MSC Scientific Deliverables

- 1°×1° multi-channel calibrated differential proton flux atlas (primary science product)
- Per-channel uncertainty maps at 1° resolution
- radiation-model residual maps per energy channel
- East-west asymmetry map with statistical significance per cell
- Monthly sub-maps (12–14 monthly averages) for temporal trend analysis
- Full public data release: calibrated counts, flux products, metadata, and uncertainty estimates

1.4.3 Extended Mission Success Criteria (EMSC)

Table 7: EMSC mission duration targets

Parameter	Value
Required duration	18 months minimum - 20 months nominal
Orbital count	8100 orbits (18 months) to 9000 orbits (20 months)
SAA transit count	3200 to 3600 SAA transits
Minimum SAA passes for EMSC science	$\geq$ 2400 passes with payload operational

Table 8: EMSC spatial mapping requirements

Parameter	Value
Grid resolution	0.5° × 0.5° in geographic latitude and longitude
Total grid cells in SAA bounding box	260 × 120 = 31,200 cells
Cell dwell time per pass	~7 seconds per 0.5° latitude band at orbital velocity

Table 9: EMSC science requirements

ID	Requirement
EM-M-01 REQUIRED	The mission shall produce a calibrated 0.5°×0.5° flux map. Primary science map includes all cells where σ_F/F ≤ 15%.
EM-M-02 REQUIRED	≥95% coverage of cells where AP9 predicts flux exceeding 1% of the mission peak flux.
EM-M-03 REQUIRED	σ_F/F ≤ 10% in at least one energy channel per cell, and ≤ 15% in all other operational channels.
EM-M-04 REQUIRED	Minimum of 5 independent SAA passes per primary-map cell.
EM-M-05 REQUIRED	0.5° map produced for each operational energy channel independently; minimum 3 channels.
EM-M-06 REQUIRED	Final 0.5° science map produced as a time-integrated product over full EMSC window, plus two 9–10 month sub-period maps.

Table 10: Comparison of success tiers

Parameter	MMSC	MSC	EMSC
Mission Duration	4–6 months	12–14 months	18–20 months
Min SAA Passes	≥ 150	≥ 900	≥ 2400
Spatial Resolution	5° × 5°	1° × 1°	0.5° × 0.5°
Map Grid Cells	312	7,800	31,200
SAA Coverage	≥ 70% (≥90% core)	≥ 90%	≥ 95% (primary)
σ_F/F Threshold	≤ 50%	≤ 25%	≤ 10–15%
Min Passes/Cell	≥ 3	≥ 5	≥ 5
Energy Channels	≥ 2	≥ 4 (all)	3 sustained
AP9 Comparison	Basic ratio check	Full residual map (1°)	Sub-map residuals (0.5°)

1.5 Outstanding MCR Evidence

Science verification path. MMSC, MSC and EMSC are retained as committed requirements. Grid-cell occupancy, detector acceptance, energy migration and systematic uncertainty must be coupled with the chosen payload response before compliance can be claimed.

2 Mission Overview and Orbit Design

The ANTARA mission has been conceived as a dedicated scientific CubeSat mission focused on characterizing the high-energy proton radiation environment within the South Atlantic Anomaly (SAA). The spacecraft is intended to operate in a near-circular Sun-Synchronous Low Earth Orbit. The detailed orbit propagation and trade-study results are provided in Appendix A.6 and Appendix B.

2.1 Spacecraft Overview

ANTARA is a 3U CubeSat with an on-board computer, an electrical power system, an attitude determination and control system, a communications system, and the scientific payload. The initial orbit is approximately 600 km at an inclination of 97.6°. No propulsion system is planned.

The mission design targets a spacecraft mass below 3 kg. The 4 kg upper allowance is not used as the spacecraft design limit. The component mass allocation, including the custom payload, will be closed using the bill of materials. The 3U spacecraft dimensions are approximately $100 \times 100 \times 340.5$ mm.

2.2 Orbit Selection

The preliminary orbit baseline is a 600 km Sun-Synchronous Orbit with an inclination of 97.6° and an eccentricity of approximately 0.001. The latest seven-case RAAN power study reports its largest modeled mean generation at 180°; power and battery behavior depend on the insertion RAAN. ANTARA does not control the launch RAAN, so the spacecraft is being assessed across multiple RAAN cases. The approved future analysis epoch is 1 January 2029; the latest source-study CSVs must still be checked for epoch and configuration consistency.

Table 11: Preliminary orbit parameters

Parameter	Mission baseline / current status
Initial altitude	Approximately 600 km
Inclination	97.6°
Eccentricity	Approximately 0.001
RAAN	Launch-dependent; 0° and 30° are favorable cases, not requirements
Propulsion	None

The mission targets approximately five years of operational science. The newest long-term drag trade predicts approximately 8.2–10.0 years to its integration endpoint, generally around 300 km, under the assumptions of the supplied runs. One run stops above 300 km, and atmospheric reentry is not simulated. These are preliminary predictions, *not* a demonstrated five-year deorbit lifetime or post-mission compliance assessment. Appendix H reports the detailed results and caveats.

2.3 Mission Lifecycle

The mission is divided into Launch, Deployment, Cold and Dark, Detumbling, Commissioning, Nominal Operations, Extended Operations, and Mission End phases. Following deployment the spacecraft activates after the planned delay, enters Cold and Dark, performs detumbling,

and proceeds toward power-positive operations and commissioning. Nominal operations consist of Sun Pointing, Detection, and Relay, with Safe Mode for recovery.

The extended phase begins after nominal mission objectives have been met. The minimum, full, and extended mission duration targets are defined in Section 1.4. The detailed phase descriptions and mode transitions are retained in Appendices B and A.1.

2.4 Design Baseline and Open Decisions

The orbit parameters above and the subsystem configurations in Section 3 are the current MCR baseline. The spacecraft design maximum is 3 kg and the approved analysis epoch is 1 January 2029. The launch RAAN is uncontrolled; final component mass allocations, payload dimensions and mission-consistent simulation configurations require confirmation. Where different simulation studies use different inputs, their results will remain associated with those inputs rather than being combined as one common spacecraft configuration.

2.5 Outstanding MCR Evidence

Orbit-study maturity. The approved future analysis epoch is 1 January 2029. Historical-epoch orbit/RAAN outputs remain provisional engineering studies until regenerated. The five-year operational target is distinct from the new preliminary 8–10-year drag-to-cutoff predictions; neither a reentry date nor disposal compliance is verified. The previously inconsistent SAA-only-orbit statistic is withheld until its counter definition is repaired.

3 Spacecraft Architecture and Ground Segment

The ANTARA spacecraft is a 3U CubeSat with the on-board computer as the main command and operations unit. The subsystem design currently uses preliminary COTS components along with the in-house scientific payload. Detailed specifications, operating estimates, and the original component tables are provided in Appendix C.

3.1 Electrical Power System (EPS)

The spacecraft uses one fixed solar panel on the +Z face and two panels stowed on the -Y face that deploy toward the +Z direction. The mission power studies use a 25 Wh battery planning capacity. The proposed ISISPACE ICEPS2 Type A has a manufacturer-rated 28.8 Wh battery-pack configuration; the mission simulation is assessed at 25 Wh. The original RAAN trade study reports average generated power of 11.05 W and 11.40 W for 0° and 30° RAAN respectively. These are study-case predictions, not a guarantee at any launch RAAN. The power studies were performed using end-of-life assumptions; their margins require acceptance against an agreed common run baseline.

The EOL power simulations provide a provisional sizing basis. Solar active area, conversion losses and worst-case EOL margin must be explicitly tied to an approved simulation configuration before these results are reported as closed budgets (Appendix A.5).

3.2 Communications System (COMMS)

The preliminary communications system uses the GomSpace NanoCom AX100 and directional and omnidirectional spacecraft antennas. The planned primary ground station is at Goa; a crossed-Yagi ground antenna is being considered. Ground-station visibility is assessed using 10° nominal and 15° conservative elevation masks. A secondary station is an option if the primary station cannot support the required operations.

The final downlink rate and storage requirement cannot yet be fixed, because the payload packet format and science-data generation are not finalized. The original 15–18 kbps and simulator 19.2 kbps values are study assumptions, not a demonstrated data-return capacity. The ground-station contact case and data-rate calculations are described in Appendix A.6.

3.3 Scientific Payload

ANTARA carries an FPDU silicon–scintillator telescope with a 5–100 MeV directional differential proton-flux requirement. The current user-supplied configuration (4 November 2026) specifies five main scintillator blocks, each $6 \times 6 \times 14$ mm, and one $6 \times 6 \times 2$ mm front scintillator. Each main block has Teflon coating on two sides, stated thickness 0.11 mm; faces and thickness interpretation remain open. EJ-550 grease couples scintillator to SiPM. These are **CONFIRMED DESIGN INPUT**, not demonstrated detector response. Block arrangement, optical segmentation and channel count require a controlled CAD/ICD. The active silicon entrance and rear penetration detector are retained; the latter is not a universal veto.

The **CANDIDATE ARCHITECTURE** is scintillator–SiPM–ASIC–ADC–microcontroller or FPGA–event classification/energy binning. The shaping parameter is 20 ns; a separate stated 20 ns ASIC-to-ADC interval has unresolved meaning. A user follow-up sets the current coincidence window to 20 ns and tentatively suggests 20 ns dead time; these remain separate quantities and timing performance **REQUIRES VALIDATION**. MCU preference requires an available coincidence-processing budget comfortably greater than 250 ns with demonstrated

margin; the final choice remains open.

The **PRELIMINARY ESTIMATE** of 400 Hz peak proton count rate assumes protons only, 250 μm (0.25 mm) aluminium shielding, geometric factor 1 cm^2 sr and 100% detection efficiency. Its simulation/spectrum/calculation provenance is pending; it is neither an all-particle trigger rate nor verified electronics capacity. Existing response curves are not replaced. Appendix A.2 specifies geometry, timing, reconstruction limitations and open decisions. The required energy interval, energy resolution, classification efficiency and science-data rate remain subject to transport, calibration and electronics verification.

3.4 Attitude Determination and Control System (ADCS)

ANTARA intends to use an integrated reaction-wheel and magnetorquer attitude-control system; the earlier CMG description is superseded. The supplied TensorTech ADCS-10m datasheet describes a CMG-equipped unit and therefore does not confirm the intended reaction-wheel plus magnetorquer architecture. The ADCS vendor/product selection remains open. The main attitude functions are initial detumbling, Sun Pointing, payload +X pointing during SAA operations, +Y relay pointing toward the ground station, and Safe Mode recovery. Manufacturer limits and exact actuator configuration will be entered after the supplier data are reconciled with the selected model. The preliminary pointing goal remains subject to simulated and hardware verification. Mode transitions are given in Appendix A.1.

3.5 On-Board Computer (OBC)

The source design uses the GomSpace NanoMind A3200 as the preliminary on-board computer. The OBC is responsible for mission sequencing, spacecraft state monitoring, command execution, payload control, data handling, and communications scheduling. The source document gives approximately 0.17 W as the average OBC power consumption for its stated operating conditions.

3.6 Spacecraft Coordinate System

The spacecraft body frame uses +X as the RAM face and payload boresight, +Y as the directional antenna face, +Z as the fixed solar-panel face, -Y as the deployable-panel packed face, -Z as the omnidirectional communications antenna face, and -X as the deployment face.

3.7 Thermal and Structural Configuration

Thermal and structural analyses are part of the MCR work, but the provided document does not yet establish qualified spacecraft temperature limits, verified mechanical modes, or completed environmental qualification. The supporting model sections are reserved in Appendix A.4; their conclusions will be included here when validated results are available.

3.8 Ground Segment and Mission Operations

The primary ground station is planned at the Goa campus, approximately 15°N, 74°E. A second station is a contingency only if the primary station cannot meet operational requirements. The ground segment provides tracking, telemetry reception, commanding, and science-data retrieval. The operational sequence uses Cold and Dark, Detumble, Sun Pointing, Detection, Relay, and Safe Mode, with Slewing as an attitude-transition state in the simulation. The full state-machine logic is given in Appendix A.1.

3.9 Outstanding MCR Evidence

Design maturity. The MCR retains the reaction-wheel/magnetorquer architecture, selected solar-panel geometry and preliminary RF concept as a feasible spacecraft configuration. Detector response, finalized ADCS supplier interface, RF throughput and structural/thermal component compliance remain explicit PDR verification tasks rather than unspecified blanks.

4 Integrated Mission Feasibility

The ANTARA simulation work combines orbit motion, power generation and consumption, attitude control, mode transitions, SAA operations, and ground-station contacts. This section gives the main results supported by the supplied historical study files. The detailed simulation descriptions, tables, and plots are retained in Appendices D and A.3.

4.1 Day-in-the-Life Operations

The reported 24-hour DITL run completes startup detumbling and repeated Detection and Relay operations. Sun Pointing occupies approximately 87.5% of that run, with Detection occupying approximately 7.3% and Relay approximately 1.0%. The run includes 18 slew events. These results apply to the source run and must not be presented as the nominal-day distribution without excluding startup operations.

The run reports six SAA passes, five of which support active proton science operations. The mean science-active fraction across those five passes is approximately 62.7%. The minimum recorded battery energy is 15.15 Wh. For the simulation's 25 Wh planning capacity this is approximately 60.6% state of charge. The uploaded 24-hour telemetry and event CSVs support the stated startup duration, contact counts, mode fractions and slews. They are based on an older mission epoch. The underlying event and telemetry statistics are documented in Appendix G.

4.2 Ground Station and Data Return

The uploaded one-day contact file records two Goa contacts with a combined duration of approximately 782 s. At a nominal 15 kbps rate this corresponds to approximately 11.73 Mbit of ideal gross downlink capacity before overheads and losses. This is geometric capacity only; final link closure is pending payload data volume, coding/modulation, link losses and actual contact scheduling. Supporting analysis is in Appendix A.6.

4.3 Updated RAAN Trade and Long-Duration Study

The latest supplied RAAN campaign evaluates seven insertion cases from 0° to 180° in 30° increments, with two runs per case. Average generated power varies from approximately 9.02 W (120°) to 11.17 W (180°), and tabulated minimum battery SOC varies from approximately 54.4% to 74.7%. These are *study-case outputs*, not a statement that the spacecraft can choose its insertion RAAN. Orbit classification totals are pooled across the two long-duration runs; they must not be mistaken for a one-year orbit tally. The full results, raw-count interpretation and drag-study limitations appear in Appendix H.

The drag CSV gives modeled termination times of roughly 8.2–10.0 years, generally near 300 km altitude. One case ends at 334.93 km, and none models final atmospheric reentry. The five-year *science/operational objective* is therefore distinct from the provisional passive-decay estimate. The campaign configuration and epoch remain to be reconciled with the approved 1 January 2029, 3 kg mission baseline.

4.4 Limitations of Current Evidence

The results in this section are simulation predictions, not hardware qualification results. The source model uses an approximate proton-flux baseline through an AP9-named interface; it

does not establish a live AP9 calculation or absolute detector calibration. Thermal, structural, payload-response, and full-year integrated outputs require validation with the corresponding source settings and result files before final acceptance.

4.5 Science Coverage Studies

Separate grid-based studies supplied with the MCR inputs examine 4–6 month, 12–14 month, and extended-duration science-map cases. Their current cell-disposition files cover the specific modeled geographic domain rather than the entire success-criteria bounding box; the resulting fractions therefore cannot directly establish the MMSC, MSC or EMSC global coverage thresholds. The evaluated cell counts and their status are tabulated in Appendix G. The detector count rate and background assumptions remain preliminary.

4.6 Outstanding MCR Evidence

Evidence qualification. Latest supplied simulation outputs are the provisional MCR reference. Each cited case must be tagged by its original epoch, spacecraft configuration and model assumptions; a 2029 rerun will supersede the historical-epoch studies. Available event logs support analysis of startup, contact opportunities and detector activity, but final science-enabled time and RF volume depend on the revised payload response.

5 Feasibility, Risk and Engineering Budgets

5.1 Mission Feasibility

The present MCR analysis supports the feasibility of a 3U spacecraft operating in a 600 km Sun-Synchronous Orbit with planned SAA detection and ground-station relay operations under the assumptions used in the source simulations. Final confirmation requires a common mass, power, payload, communications and ADCS baseline and the corresponding verified simulation results.

5.2 Measures of Performance and Effectiveness

The draft defines Measures of Performance (MOPs) for battery state of charge, Detection Mode pointing accuracy, science-data return, payload livetime and mission lifetime. Measures of Effectiveness (MOEs) address verified proton measurements, data return, radiation mapping and radiation-model comparison. Their original thresholds and goals are retained in Appendix E pending reconciliation with the final spacecraft configuration.

5.3 Engineering Budgets

The supplied bill of materials totals 1536.3 g for the rows with specified masses; this is not a complete spacecraft mass because the payload and deployment hardware are unallocated and the panel assembly quantity still requires checking. A design mass below 3 kg is targeted. Component costs are currently marked TBD. The payload and deployment hardware are not yet fully included in the mass total, and several component costs are still to be determined. The power analysis uses 25 Wh as the mission battery planning capacity; the proposed Type A datasheet gives 28.8 Wh rated. Final science-data rate and communications closure depend on payload decisions. The detailed component tables are retained in Appendix E, with reserved consolidated budgets in Appendix A.5.

5.4 Risks and Requirements

The draft identifies deployment failure, detumbling failure, loss of pointing, payload radiation damage, loss of communications, battery degradation and thermal excursion as mission risks. The original risk register does not yet provide complete likelihood and consequence scores. The requirements traceability matrix is preliminary and will need to be updated against the reorganized document and the final design values. Both source tables remain in Appendix E.

5.5 End-of-Life and Debris Mitigation

ANTARA does not carry propulsion and therefore relies on passive orbital decay for disposal. Its working operational science objective is approximately five years. The newly supplied long-term orbital-drag CSV reports modeled termination after approximately 8.2–10.0 years across seven RAAN cases and two runs per case; most traces end near 300 km, while one ends at approximately 335 km. This is a numerical propagation endpoint, not physical atmospheric reentry. A validated launch-epoch-specific drag model, final spacecraft aerodynamic properties, and subsequent atmospheric-decay assessment are required before making disposal-compliance claims. The full study is presented in Appendix H.

5.6 MCR Open Items

Before the next review, the team must confirm the spacecraft mass and component baseline; reconcile the EPS energy capacity and electrical loads; finalize the ADCS actuator model and FPDU dimensions; validate the link and detector count assumptions; and connect each accepted simulation result to its source configuration. The supporting technical material is organized in the appendices.

5.7 MCR Outcome and PDR Scope

The MCR establishes a preliminary mission baseline and examines the conditions under which the mission can be feasible using proposed COTS components and a custom payload. The final subsystem data paths, control paths, power distribution, communication implementation and payload design will be developed for the Preliminary Design Review (PDR). The present MCR should not be interpreted as completed hardware qualification or final selection of components whose interfaces and performance have not yet been verified.

5.8 Outstanding MCR Evidence

MCR decision and next review. The MCR defines a conditionally feasible spacecraft concept with a maximum design mass of 3 kg. The detailed payload-electronics, data, ADCS control, power-distribution and RF interfaces are PDR design products. PDR entry requires traceable subsystem requirements, reconciled BOM, validation of the updated 8–10-year drag-to-cutoff study and a separate disposal assessment and measured or source-backed margins.

A Detailed Technical Supporting Analysis

A.1 Detailed Operational Mode Transition Logic

The ANTARA concept of operations uses explicit spacecraft modes, bounded transition guards and a safety override. Figures 2 and 3 show the normal mission sequence and supervisory logic. The exact software thresholds are configuration-controlled simulation parameters, rather than independently qualified flight-software settings.

A.1.1 Operational Modes and Attitude Targets

Table 12: Mode functions and commanding intent

Mode	Attitude/control goal	Subsystem behavior and exit condition
Cold & Dark	Deployment initialization; no operational pointing	Minimal EPS/OBC; initialize health monitoring and release startup delay; then Detumble.
Detumble	Reduce body rates using magnetic control	EPS/OBC/ADCS active; exit when rate and stability guard pass; otherwise remain or enter Safe on faults.
Sun Pointing	+Z solar face directed toward Sun	Default power-recovery and power-positive standby; permit scheduled SAA science or contact after safety guards.
Science / Detection	+X payload boresight aligned with required science pointing	Read out FPDU during a permitted SAA opportunity; account for payload warm-up, slew and available energy.
Relay / Downlink	+Y directional antenna aimed toward selected ground station	Radio TX/RX and queued-data transfer during usable elevation window; recover normal pointing after contact.
Passive / Eclipse	Preserve previous attitude during eclipse	Minimize active Sun-search maneuvers; resume Sun pointing or the applicable scheduled activity when illumination and guards permit.
Slewing	Move between required attitude targets	Temporary ADCS phase; next operational mode begins after reorientation/settling; abort on fault or low power.
Safe	Survival, battery conservation and anomaly handling	Suspend nonessential loads; maintain health, available communications and recovery control. Re-enter nominal path only after recovery guards.

A.1.2 Nominal Startup and Science/Relay Sequence

A.1.3 Supervisory Priority and Safety Overrides

A.1.4 Transition and Guard Matrix

Table 13: Required operational transitions and relevant checks

From	To	Guard / interpretation
Cold & Dark	Detumble	Boot/commissioning delay complete, EPS and OBC alive.
Detumble	Sun Pointing	Body rate falls below accepted rate threshold and attitude solution remains stable.

From	To	Guard / interpretation
Sun Pointing	Slewing → Science	SAA opportunity exists, safe battery reserve and payload readiness; +X boresight target feasible.
Sun Pointing	Slewing → Relay	Ground contact above selected mask, queued data or command requirement, sufficient energy; +Y antenna acquisition.
Science	Sun Pointing / Relay	SAA window ends or a higher-priority valid contact is scheduled; use Slewing as needed.
Relay	Sun Pointing / Science	Contact ends, queue complete or higher-priority science becomes available; attitude guard applies.
Sun Pointing	Passive / Eclipse	Eclipse begins; passively retain last attitude rather than repeatedly Sun-searching.
Passive / Eclipse	Sun Pointing	Sun visibility returns and recovery/scheduling permits active pointing.
Any operational mode	Safe	Critical battery, ADCS, thermal or spacecraft health condition supersedes nominal task.
Safe	Detumble / Sun Pointing	Fault clearance and recovery guards satisfied; select Detumble if rates require stabilization.
Slewing	Scheduled target / Safe	Target within pointing/settling tolerance or slew interruption on fault/resource guard.

A.1.5 Timing, Power and Science Opportunity Accounting

A useful SAA crossing is not automatically a valid measurement window. The data-taking interval is the portion of the crossing for which the payload is enabled, pointing has settled, coincidence acquisition is active and no resource/safety override applies. Equivalently,

$$T_{\text{science,usable}} = \int_{t_0}^{t_1} \mathbf{1}_{\text{SAA}}\, \mathbf{1}_{\text{payload}}\, \mathbf{1}_{\text{pointing}}\, \mathbf{1}_{\text{health}}\, dt. \tag{1}$$

The integrated logs distinguish transitions, slews, SAA passages, accepted proton events and ground-station contacts. The current one-year provisional output reports 2,346 SAA-passage records and 820 primary-station contact records. These are *event-log counts for the supplied historical-epoch run*, not unique orbital revolutions or verified 2029 yields. A 2,460-second startup detumble event is observed in that run. These values are to be reconciled with the approved 2029 run before final MCR exit.

A.1.6 Implementation Boundary

The mode architecture, fault-preemption policy, Sun-pointing fallback and science/relay sequencing are MCR-level decisions. Exact battery-voltage thresholds, rate limits, thermal recovery timers, attitude-control gains, antenna acquisition tolerances and software task deadlines belong to the controlled ADCS/OBC configuration and must be verified at PDR. This section intentionally describes the complete concept and its verification boundary without claiming an untested flight implementation.

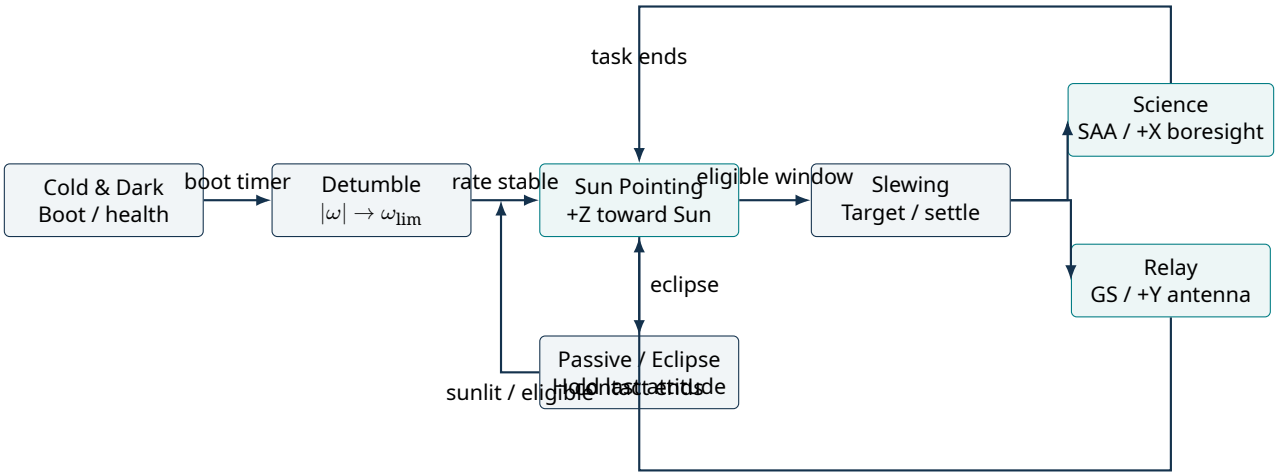


Figure 2: Information-bearing functional flow of the planned mission sequence. Windows and guards come from the mission scheduler. The schematic does not assert that every transition is immediate; slew and settling times are accounted for separately.

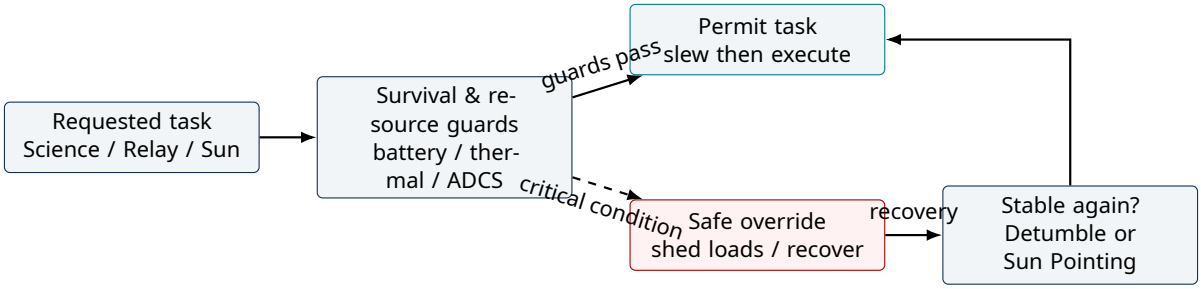


Figure 3: Supervisory decision logic: survival constraints override scheduled science and downlink. Recovery transitions depend on the fault and attitude state; this is a mission-level policy, not an assertion of finalized flight code.

A.2 Detailed Scientific Payload: FPDU Current Design Inputs

The FPDU retains the mission requirement to measure directional differential proton flux over 5–100 MeV. This appendix is the current payload narrative specification, incorporating authoritative user inputs dated 4 November 2026 as configuration `FPDU-2026-11-04`. It updates the design description in *ANTARA FPDU Payload Redesign, Rev. 2.1* (18 September 2026); it does not imply approval of a new numbered hardware revision. Neither the new geometry nor the proposed electronics has demonstrated the required energy response.

Status convention. **CONFIRMED DESIGN INPUT** identifies supplied dimensions or construction choices, not measured performance. **PRELIMINARY ESTIMATE** identifies the conditional proton-rate estimate. **CANDIDATE ARCHITECTURE** identifies unselected electronics and the processor trade. **REQUIRES VALIDATION** applies to energy response, calibrated efficiency, optics, timing margin and event classification.

A.2.1 Current Geometry and Retained Interfaces

Table 14: Current FPDU geometry inputs and retained detector concepts

Element	Supplied specification	Interpretation / open interface
Main scintillator	Five blocks, each $6 \times 6 \times 14$ mm	Material dimensions; arrangement, axis orientation and segmentation unresolved
Main-block reflector	Teflon, two sides of each block; stated thickness 0.11 mm	Coated faces and per-face versus total thickness unresolved
Front scintillator	One, $6 \times 6 \times 2$ mm	Position relative to silicon and readout unresolved
Optical interface	EJ-550 coupling grease	SiPM-to-scintillator coupling; thickness and geometry unspecified
Active silicon entrance	Retained from Rev. 2.1	Thickness, active area and interface to front scintillator require reconciliation
Rear active detector	Retained penetration tag	Not an unconditional veto; geometry and readout unresolved
Al entrance / collimator	Retained concept	Flight thickness and acceptance still require transport/geometry verification

The first four rows are **CONFIRMED DESIGN INPUT**. Scintillator composition is not newly specified; EJ-200 remains historical candidate evidence, not a confirmed material assignment. Five physical blocks are not necessarily five independently instrumented layers or stopping stages. Rev. 2.1 described one main block with two-side minimum/four-side optional SiPM readout. Whether the five blocks form one optical assembly is unresolved; neither that interpretation nor a readout-channel count is adopted. No current FPDU assembly CAD/ICD was found in the supplied component archive.

Nominal scintillator material volumes are $6 \cdot 6 \cdot 14 = 504$ mm^3 (0.504 cm^3) per main block, 2520 mm^3 (2.520 cm^3) for all five, and 72 mm^3 (0.072 cm^3) for the front piece. Total scintillator material volume is 2592 mm^3 (2.592 cm^3). These exclude Teflon, grease, gaps, SiPMs, silicon, rear detector, housing and supports. External packaged dimensions and mass cannot be derived from these inputs.

A.2.2 Dimensioned Concept and Optical Coupling

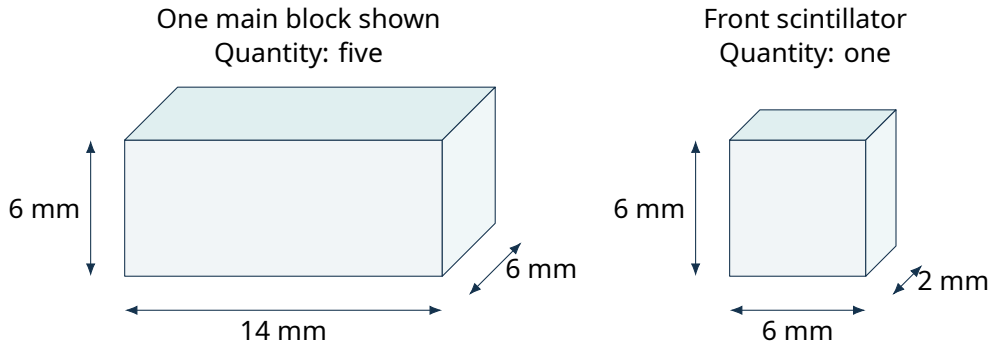


Figure 4: Dimensioned component concepts, not an assembly drawing and not to a common scale. Dimensions describe scintillator material. Teflon coats two sides of each main block with stated thickness 0.11 mm; faces and thickness interpretation are unresolved and are deliberately not drawn. Drawing axes do not establish flight orientation.

SiPMs are optically coupled to scintillator using **EJ-550 optical coupling grease** (**CONFIRMED DESIGN INPUT**). No grease thickness, transmission efficiency, refractive-index matching performance or coupling geometry is specified. Surface finish, optical segmentation, reflector placement, SiPM count, photon-detection efficiency, gain, saturation, quenching and temperature dependence are marked **REQUIRES VALIDATION**; no optical response curve is assigned.

The inherited Rev. 2.1 entrance concept used a 10 mm aperture radius, 30 mm spacing and a 10×10 mm active silicon area, yielding the ideal sharp-cone estimate $G = A\pi \sin^2[\arctan(10/30)] \simeq 0.314$ cm^2 sr. This is historical geometry evidence, not acceptance for the supplied blocks. It differs from the conditional rate assumption in Section A.2.5; neither value establishes the revised detector's calibrated effective geometric factor.

A.2.3 Proposed Electronics Signal Chain

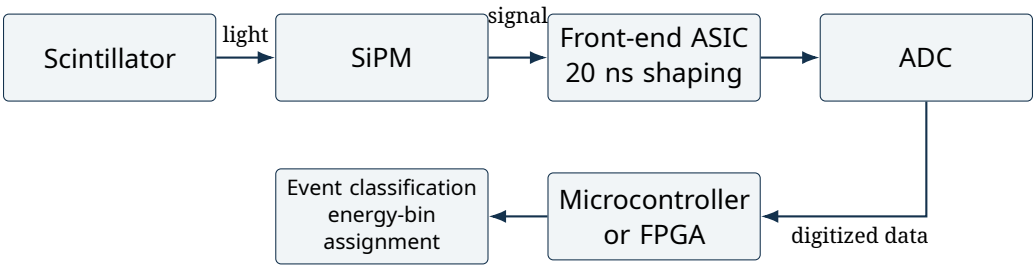


Figure 5: CANDIDATE ARCHITECTURE. EJ-550 couples scintillator to SiPM. A further 20 ns was stated between ASIC output and ADC processing; its meaning is unresolved. This does not establish a 40 ns total latency. Silicon and rear-detector front ends and their timing interfaces still need an ICD.

The SiPM converts scintillation light to an electrical signal routed through ASIC and ADC. Digitized information reaches a microcontroller or FPGA for event classification and energy-bin assignment. No ASIC/ADC/processor model, ADC sampling rate or resolution, trigger topology, buffer design or throughput has been selected here. Rev. 2.1's FPGA-only language is superseded by this conditional processor trade.

A.2.4 Timing Budget and Processor Selection

Table 15: Separate timing and capacity terms; no completed timing closure

Quantity	Current input	Required evidence / interpretation
Coincidence window width	20 ns	Current user follow-up input; reference edges and inclusive boundary to define
Timing resolution and jitter	Unspecified	Timestamp/discriminator resolution and channel skew measurements
Analog shaping parameter	20 ns	Supplied ASIC parameter; pulse definition and selected device needed
ASIC output to ADC interval	20 ns stated	Meaning unresolved: delay, acquisition, conversion or other interval
ADC acquisition/conversion	Unspecified	Datasheet and measured pipeline behavior; not inferred from shaping
Digital processing latency	Unspecified	Worst-case classification/binning latency for candidate processor
Sustained / peak throughput	Unspecified	All-particle triggers, rejection load and ADC service capacity
Readout dead time	20 ns tentative	User qualified this as a guess; measured value and shared/per-channel topology unresolved
Buffering / loss probability	Unspecified	Queue depth, burst tests, overflow counters and live-time accounting

The 20 ns coincidence width is a **CONFIRMED DESIGN INPUT**, not measured timing performance. The tentative 20 ns dead time is recorded separately as **REQUIRES VALIDATION**; it is not adopted as a qualified correction. Equal numerical values do not make coincidence width, shaping time, ADC interval and dead time the same physical quantity, nor show that they are independent serial delays.

CANDIDATE ARCHITECTURE: prefer a microcontroller if the available coincidence-processing timing budget is comfortably greater than 250 ns, with demonstrated engineering margin. Consider an FPGA when a microcontroller cannot meet the timing constraints. The available processing budget is not automatically the 20 ns coincidence window: hardware timestamps, buffering and acquisition architecture may separate these. Conversely, if logic must complete within that window, the stated MCU preference condition is not met. The final decision needs a defined latency path, worst-case workload, jitter/skew allocation, a configurable agreed margin and measured throughput; no MCU or FPGA feasibility conclusion is made.

A.2.5 Conditional Proton Rate and Data Product

PRELIMINARY ESTIMATE: the current estimated peak proton count rate is **400 Hz**, conditional on **protons only**, **250 μm (0.25 mm) aluminium shielding**, an **assumed geometric factor of 1 cm^2 sr**, and **100% assumed detection efficiency**. This is user-supplied; its generating simulation, input spectrum and calculation have not been identified. Energy integration bounds, angular distribution, environment epoch and location/percentile are also unspecified. It is not a verified maximum electronics event rate, not an all-particle trigger rate, and not proof of instrument performance. The shielding value is a rate-estimate assumption, not a validated flight entrance-thickness selection.

The geometric factor includes solid-angle acceptance and must not be treated as area alone.

Existing differential-flux, integrated incident-rate, expected accepted-count and electronics-trigger quantities remain distinct. No existing rate curve is replaced by a constant. Historical 100 Hz/500 kHz estimates and the separate pile-up study's assumed 100 kHz remain superseded or study-specific evidence, not revised flight limits.

Histogramming with diagnostic background, saturation, overflow and live-time counters remains the intended product. A planning array of 64 four-byte counters occupies 256 bytes before headers, diagnostics and cadence; this is neither a demonstrated energy resolution nor a packet-rate prediction. Payload power and actual science telemetry require the selected electronics and packet definition. Existing EPS and telemetry outputs are unchanged study results.

A.2.6 Event Reconstruction and Validation Limits

REQUIRES VALIDATION. The retained event concept combines active-silicon energy deposition, scintillator signals, rear-penetration information, timestamps and saturation/quality flags. Front-piece and per-block observables depend on the unresolved optical/readout topology. A silicon–main coincidence is a candidate trigger; the role of the new front scintillator is not established. Rear activity must be retained for penetrating science events rather than universally vetoed. Stopping and penetrating reconstruction functions and a mixed-species confusion matrix must be calibrated before assigning physical energy bins. No legacy stopping thresholds or efficiencies transfer to this geometry.

For differential directional incident flux Φ_s in particles cm^{-2} s^{-1} sr^{-1} MeV^{-1}, define the directional effective response $A_{j,s}$ in cm^2, including acceptance and migration into category j. Then expected counts over an exposure are

$$N_j = \int_{t_0}^{t_1} \ell(t) \sum_s \int\int A_{j,s}(E,\Omega,t)\, \Phi_s(E,\Omega,t)\, dE\, d\Omega\, dt, \tag{2}$$

where ℓ is a validated live fraction (or a channel-dependent treatment if required). Its time derivative is a rate, not a count; an angle-integrated geometric factor must not be multiplied by a second solid-angle integral. Background, calibration and migration uncertainty must accompany counting statistics. The 5–100 MeV requirement and science-success targets remain; achievement is not established by completed orbit propagation or accumulated synthetic counts.

A.2.7 Required Evidence and Open Decisions

Table 16: Evidence required before revised FPDU performance claims

Decision / model	Verification needed
Assembly and optics	Controlled CAD/ICD: positions, axes, two coated faces, thickness interpretation, material, optical segmentation, SiPM channels and EJ-550 interfaces
Entrance and acceptance	Al/Si transmission sweep, front-piece placement, full 3-D angular acceptance and as-built dimensions
Energy / species response	Archived spectra; GRAS/Geant4 deposits and optical/electronics model; proton energy/angle scan and mixed-field confusion matrix
Timing and processor	Resolve ADC interval, coincidence definition, measured dead time, worst-case latency margin, saturation and burst/queue loss tests
Data and power	Defined histogram edges, diagnostic counters, overflow behavior, packet cadence and measured component power

The inherited analysis plan uses mission-epoch SPENVIS AE9/AP9 spectra, ion/background inputs, MULASSIS entrance trades and full-geometry GRAS/Geant4. No completed response for the new configuration has been recovered. The source archive scores deposits in old layers; it does not implement the new optics, ASIC, ADC or classifier. A versioned declarative configuration is therefore provided without a fabricated physics adapter. Detailed decisions and compatibility dispositions are recorded in the payload audit follow-up documents.

A.2.8 Superseded Evidence

The historical *Payload_1* and DITL/Geant4 models use five stopping layers of 2, 4, 12, 22 and 30 mm. These are explicitly superseded evidence, not the supplied five $6 \times 6 \times 14$ mm blocks. Rev. 2.1's single-block description is retained as design history; its relationship to a five-block optical assembly must be clarified. Silicon and rear-detector concepts remain. Existing numerical outputs are preserved, but none qualifies this revised geometry, its 5–100 MeV response or electronics feasibility.

A.3 Simulation Configuration and Provenance

This appendix records the simulation code used to support the ANTARA mission concept. The simulation scripts are organised by study under `src/simulations/`. The integrated science simulation uses a common configuration defined in `src/simulations/helper/antara_models.py`; the standalone orbit, power, pointing, detumble, CONOPS and thermal studies have their own study-specific configurations. Their results must not be treated as outputs of one identical spacecraft model unless the input parameters have been checked.

A.3.1 Simulation Inventory

Table 17: Simulation source files and principal applications.

Study	Python entrypoint(s)	Intended evidence
Integrated science and DITL	`ditl/ANTARA_Science_Sim.py`; `ditl/scenarios.py`; `ditl/compare_scenarios.py`	Mission modes, power, attitude, SAA science opportunities and ground contact
Orbital decay	`orbit/ANTARA_drag_finalconfig.py`; `orbit/ANTARA_drag_study.py`	Orbital decay and lifetime studies
Power	`power/ANTARA_MCR.py`; `power/ANTARA_avg.py`; `power/ANTARA_FINAL_POWERGEN.py`; `power/TradeCase.py`	Power-generation and battery trade studies
Pointing	`pointing/ANTARA_pointing_budget.py`	Ground-station pointing geometry
Detumble	`detumble/leop_detumble_mc.py`; `detumble/analyze_detumble_mc.py`	Initial-rate reduction and Monte Carlo statistics
CONOPS	`conops/ANTARA_conops.py`	SAA and station encounter analysis
Thermal	`thermal/ThermalSimulator.py`	Lumped-parameter temperature studies
Shared support	`helper/antara_models.py`; `helper/fsm_spec.py`; `helper/telemetry_metrics.py`; other `helper/` modules	Configuration, attitude models, FSM rules and telemetry processing
Historical	`legacy/ANTARA_Master.py`	Superseded simulation; not a current baseline

A.3.2 Integrated Simulation Baseline

The values in Table 18 are the defaults in `MissionConfig` supplied with the reviewed source archive. These are simulation inputs, not final hardware qualifications or independent performance measurements. A command-line override or a run-specific configuration can change them.

Table 18: Selected integrated simulation configuration defaults.

Parameter	Code-defined value	Unit
Initial orbital altitude	600	km
Inclination	97.6	degrees
Eccentricity	0.001	—
RAAN	15	degrees
Initial spacecraft mass	3.0	kg
Drag coefficient	2.2	—

Parameter	Code-defined value	Unit
SRP coefficient	1.8	—
Simulation start date	1 March 2000	UTC calendar date
Nominal fine time step	2	s
Coarse time step	20	s
Battery planning capacity	25	Wh
Initial stored battery energy	18	Wh
EPS charge efficiency	0.88	—
Solar temperature derating	0.88	—
Maximum solar generation parameter	22	W
OBC load parameter	0.8	W
EPS load parameter	1.2	W
Payload load parameter	10	W
Radio transmit load parameter	12	W
ADCS idle and slew loads	0.8 / 4.5	W
Radio downlink rate	0.0192	Mbit/s
Primary ground-station location	15° N, 74° E	coordinates
Ground-station altitude	70	m
Ground-station elevation mask	15	degrees
Random seed	42	—

The integrated model also defines power thresholds for entering and exiting low-power conditions, attitude-pointing limits, detection and relay minimum durations, a maximum wheel-saturation hold, and fault/recovery debounce settings. The exact configuration used for a reported run must be retained with that run, rather than inferred only from current defaults.

A.3.3 Integrated DITL Scenarios

The three standard presets are defined in `ditl/scenarios.py` and shown in Table 19. The two 24-hour cases separate the startup sequence from nominal operations; the one-year preset includes startup.

Table 19: Integrated Science simulator presets.

Preset	Duration	Starting condition
`leop_24h`	24 h	Includes cold-start and detumble sequencing
`no_leop_24h`	24 h	Skips LEOP and starts in Sun Pointing
`leop_1yr`	8,760 h	Includes LEOP; uses multirate integration with a configured 20 s coarse step

Configuration precedence in the source is explicit command-line options, followed by scenario preset, followed by the global defaults. In particular, the RAAN and battery capacity are configuration fields and are not exposed as direct `--raan-deg` or `--battery-wh` command-line options in the documented Science entrypoint.

A.3.4 Numerical and Physics Models

The integrated simulator uses Orekit numerical orbit propagation and includes gravitational perturbations, an atmospheric drag model and solar radiation pressure. The source im-

ports the Holmes–Featherstone gravity-field model and the DTM2000 and NRLMSISE-00 atmospheric models. The model and options active in a particular run must be established from the saved configuration and executed code path.

The integrated model uses scalar-first quaternions, $[w, x, y, z]$, for body-to-inertial attitude transformations. The spacecraft body axes are defined with $+X$ along the payload boresight, $+Y$ toward the relay antenna and $+Z$ along the fixed solar-panel face. The simulation includes finite slews and pointing-settlement checks; an attitude command does not produce an instantaneous orientation change.

The magnetic-field implementation supports the configured dipole option and an IGRF backend through `ppigrf`. If the optional IGRF package is unavailable, the documented IGRF path warns and falls back to a centred dipole. This fallback is not equivalent to an IGRF-based result and must be recorded when used.

The proton environment model applies SAA spatial variation, magnetic pitch-angle dependence, east–west asymmetry, storm scaling and detector response to a configured baseline spectrum. Although a legacy interface and some telemetry keys use the name ap9, the present source does *not* execute a live AP9 calculation. Consequently, simulator flux values must not be represented as independently generated AP9 predictions or validated in-orbit measurements. The simulator's detector counts are modeled expected counts, rather than a calibrated hardware measurement campaign.

A.3.5 Time Accounting and Data Outputs

The Science simulator writes scenario-tagged CSV files under `results/science/`, including telemetry, SAA passes, ground-station contacts, mode transitions, slew events, detumble events, rejected transitions and proton passes. Figures are written under `figures/science/` when plot generation is enabled.

Telemetry distinguishes the row observation time, `time_sec`, from the integrated interval end, `state_time_sec`. `interval_duration_sec` records the elapsed integration interval. The final terminal snapshot has zero duration. Time-averaged quantities and mode occupancy must therefore be weighted by the interval durations rather than the number of CSV rows. Plot sampling is for display only; extrema and scientific statistics should be calculated from complete telemetry and validated summary outputs.

The simulator supports trusted, versioned checkpoints and resumable execution. Checkpoints and their linked science-event archives must be retained together. A resume record does not by itself establish that the entire intended scenario completed correctly.

A.3.6 Supporting Study Differences

The standalone scripts should be treated as supporting studies rather than interchangeable implementations of the integrated Science scenario. In particular:

- The standalone pointing-budget study uses its own nadir-pointing and antenna-boresight assumptions; these must be compared with the integrated $+Y$ relay antenna geometry before a result is used as the spacecraft relay pointing budget.
- Standalone solar-power and battery studies contain their own panel and operating assumptions, which must be reconciled with the integrated Science configuration.
- Detumble Monte Carlo trials are useful for testing the controller over initial conditions, but do not automatically demonstrate integrated EPS, payload and ground-station operation.

- The thermal model is a separate lumped-parameter study, not a temperature field directly solved within every Science simulator step.
- The historical `ANTARA_Master.py` is superseded and should not be used as the current simulation baseline.

A.3.7 Verification Status and Known Limits

The supplied repository's `docs/CURRENT_STATUS.md` records a 414-test software regression run during Phase 3B and matching one-hour Science baseline outputs. These are repository-reported verification results; they have not been independently reproduced in preparing this appendix. The same record states that a 60-hour annual-preset run does not establish completion or validation of the full one-year scenario. It also identifies the absence of live AP9 calculation and the superseded status of the Master implementation.

For each figure or numerical conclusion adopted in the MCR, retain the following alongside the output: source revision or commit, scenario name, complete run configuration, RNG seed where applicable, simulator start date, run duration and completion status, numerical step controls, output path, and any modifications to the baseline model. Without these details, a plotted quantity cannot be reliably associated with a particular spacecraft design configuration.

A.3.8 Simulation Evidence Record

Table 20 is to be filled as individual simulation results are approved for the MCR. Entries are deliberately not assigned values without their actual completed output files.

Table 20: Approved simulation output provenance (to be completed from run artifacts).

Approved result	Source revision / run ID	Configuration and seed	Archived data / figure
LEOP 24-hour DITL	Pending	Pending	Pending
Nominal 24-hour DITL	Pending	Pending	Pending
One-year integrated mission	Pending	Pending	Pending
Orbit and RAAN trade study	Pending	Pending	Pending
Power and battery trade	Pending	Pending	Pending
Detumble Monte Carlo	Pending	Pending	Pending
Thermal cases	Pending	Pending	Pending

Payload input update. The declarative configuration `FPDU-2026-11-04` records supplied geometry, optical coupling and separate electronics timing inputs. It is not consumed by existing DITL or Geant4 code. Their legacy five-layer response, efficiency and packet assumptions cannot qualify the updated payload. The current coincidence input is 20 ns; a separately recorded tentative 20 ns dead-time estimate is unvalidated. No existing output was recalculated or relabeled as a new-geometry result. The payload compatibility review records the missing geometry-to-response-to-packet interface.

A.4 Thermal and Structural Supporting Analysis

This appendix describes the thermal model implemented in `src/simulations/thermal/ThermalSimul` and the structural information available in the MCR. The thermal results are model predictions; no spacecraft-level thermal-vacuum or mechanical qualification is claimed.

A.4.1 Thermal Model and Spacecraft Geometry

The simulator represents the spacecraft with seven lumped-capacitance nodes: the six external faces and an internal bus. The geometric model uses a $10\times10\times34$ cm 3U body with the long axis along X. Each end face has area $0.010\,\mathrm{m}^2$ and each long face has area $0.034\,\mathrm{m}^2$. Each node carries mass, specific heat, solar absorptivity, infrared emissivity and thermal conductance to the internal bus. Neighboring external faces are also coupled through a specified conductance. These values form a preliminary thermal representation, not measured properties of the final assembly.

Table 21: Selected thermal simulation inputs (`ThermalMissionConfig` defaults).

Input	Code setting
Initial orbit altitude / inclination	600 km / 97.6°
RAAN and eccentricity	15° / 0.001
Spacecraft model mass	3.0 kg
Simulation duration / nominal time step	24 h / 2 s
Thermal substeps per simulation step	10
Initial temperature	293.15 K
Internal bus lumped mass / specific heat	1.8 kg / 800 J kg^{-1} K^{-1}
Face-to-face thermal conductance	0.05 W K^{-1}
Solar constant / deep-space temperature	1361 W m^{-2} / 2.7 K
Applied temperature margin parameter	10°C

A.4.2 Thermal Balance and Numerical Method

For each node i, the model integrates a lumped heat balance of the form

$$C_i\frac{\mathrm{d}T_i}{\mathrm{d}t} = Q_{\mathrm{solar},i}+Q_{\mathrm{albedo},i}+Q_{\mathrm{EarthIR},i}+Q_{\mathrm{internal},i}-Q_{\mathrm{radiated},i}+\sum_j G_{ij}(T_j-T_i), \qquad C_i = m_i c_i. \tag{3}$$

The code calculates incident solar radiation, Earth albedo and infrared heating, thermal emission, internal dissipation and conduction. Solar-cell electrical conversion is included in the face heat balance. The orbit is propagated with Orekit numerical integration; the thermal equations use fourth-order Runge–Kutta integration with configurable thermal substeps. Earth albedo and infrared terms may vary with latitude in the implemented model.

A.4.3 Nominal, Hot and Cold Cases

The source defines `NOMINAL`, `HOT` and `COLD` bounding-case configurations. Nominal retains the base configuration. The HOT case places the model near perihelion, increases the end-of-life optical assumption and adds 0.5 W of bus dissipation. The COLD case uses an aphelion epoch, a lower optical assumption and 0.5 W less bus dissipation; relay and detection loads

are disabled for that case. These are deliberately selected model cases, not measured flight extremes. The simulation can also use LVLH or prescribed tumbling attitude. The default internal heat schedule is independent of the Science DITL flight software: it includes a 1.5 W baseline bus, 0.4 W charging term, 0.8 W eclipse-heater term, a 3.0 W relay-pass increment and a 1.2 W detection-window increment with configured durations. Consequently, thermal case temperatures should not be described as arising from the exact commanded state history of `ANTARA_Science_Sim.py`.

A.4.4 Temperature Outputs and Required Comparison

The source produces thermal telemetry, a worst-case summary CSV and plots of temperatures, orbital context and radiation contributions. The final temperature limits of the actual EPS, batteries, payload, OBC and ADCS should be compared with the extrema of the corresponding nodes for the selected bounding cases. The comparison remains open until approved case outputs and component limits are matched to the same configuration.

Table 22: Thermal verification items to close using run outputs and component specifications.

Item	Evidence needed	Status
HOT maximum	Maximum node temperatures and applicable upper component limits	Pending reconciliation
COLD minimum	Minimum node temperatures, heater assumptions and lower limits	Pending reconciliation
Payload temperature stability	Detector and electronics thermal range, coupling and bias correction	Preliminary
Case reproducibility	Case identifier, source revision, orbit epoch, input parameters and outputs	Required per run

A.4.5 Structural Configuration and Verification

The current MCR specifies a 3U spacecraft envelope of approximately $100 \times 100 \times 340.5$ mm, a one-fixed/two-deployable-panel configuration and a preliminary component mass table. Structural verification will require the selected structure and deployer interface specification, consolidated mass and inertia properties, solar-panel deployment design and the applicable launch mechanical environment. The provided simulation archive does not establish a completed structural finite-element, modal, vibration or shock qualification analysis. These items therefore remain planned verification work, rather than numerical results to be inserted into the report.

A.5 Detailed Engineering Budgets and Bill of Materials

This appendix records the calculation method and the presently available design inputs for the spacecraft budgets. The existing itemised mass and cost tables are retained in Appendix E. The simulation constants below must not be substituted for component datasheet values without identification.

A.5.1 Component Baseline and Mass Budget

The original draft lists the integrated ADCS, OBC, one body-mounted panel, deployable solar panels, EPS and battery, CubeSat structure, radio, directional and omnidirectional antennas, GNSS receiver and antenna, deployment interface and custom payload. Payload and some mechanism masses and several prices remain unfilled. Because the original component list is not yet a reconciled assembly-level budget, a total wet mass and center of gravity cannot be certified from it.

$$m_{\text{total}} = \sum_i n_i m_i + m_{\text{harness}} + m_{\text{mechanisms}} + m_{\text{contingency}}. \tag{4}$$

For each entry, record quantity, unit mass, selected or proposed part number, data source and whether the quantity is measured, quoted or estimated. The 4.0 kg mass used in the annual RAAN campaign and orbital-decay standalone study differs from the 3.0 kg integrated DITL and thermal defaults. Neither should be silently reported as an as-built mass.

A.5.2 Solar Generation and EPS Budget

The standalone annual RAAN study `power/ANTARA_MCR.py` uses one fixed and two deployed solar panels with fixed area 0.030 m^2, deployed area 0.030 m^2 per wing, cell efficiency 0.28 and a system efficiency factor 0.513. It assumes 8.0 W constant load and a 25 Wh battery. The separate 24-hour `ANTARA_FINAL_POWERGEN.py` instead uses a 0.021 m^2 fixed panel, two 0.018065 m^2 wings, 0.30 cell efficiency, 0.72675 system factor, nadir-pointing attitude and a 97.4° inclined orbit. Those studies are different configurations, and their output values are not interchangeable. For an illuminated panel the basic generation calculation is

$$P_{\text{array}} = S\,\eta_{\text{cell}}\,\eta_{\text{system}} \sum_k A_k \max(0, \hat{n}_k \cdot \hat{s})\, f_{\text{deg},k}, \tag{5}$$

where S is incident solar irradiance, A_k is the panel area, $\hat{n}_k$ and $\hat{s}$ are the panel normal and Sun direction, and $f_{\text{deg},k}$ is a model degradation factor where applied. Some source studies use independently tracking wings instead of the common attitude cosine, so their specific implementation must be cited.

A.5.3 Battery Sizing and Depth of Discharge

The analytical `power/TradeCase.py` compares capacities of 10, 15, 20, 25 and 30 Wh with a fixed 97-minute period, 8 W load and a precomputed eclipse-fraction table. Its eclipse discharge estimate is

$$\text{DoD} = \frac{P_{\text{load}}\, f_{\text{eclipse}}\, T_{\text{orbit}}}{E_{\text{battery}}}. \tag{6}$$

The companion state-of-charge illustration uses 10-second steps and an assumed sunlight recharge margin of 1.3 W above the load. The study's hardcoded cycle-life figures are assumptions, not a demonstrated battery qualification. The integrated DITL uses a 25 Wh planning capacity, while the MCR also names an ICEPS2 Type A product with a 22.5 Wh specification and another table refers to 28.8 Wh. Final energy margin must be recalculated using the selected battery's usable beginning- and end-of-life capacity.

A.5.4 Mode-Resolved Electrical Loads

The main MCR gives preliminary mode-power ranges, while the integrated Science code defines its own component-specific loads, including OBC 0.8 W, EPS 1.2 W, payload 10 W and radio TX 12 W. These code inputs should not be mistaken for measured flight hardware loads. For each mode, the detailed budget should sum simultaneous loads and distinguish peak power from time-averaged energy:

$$\Delta E = \int_{t_0}^{t_1} \left(P_{\text{charge}}(t) - P_{\text{loads}}(t) \right) \, \mathrm{d}t. \tag{7}$$

The expected mode timeline, attitude-related power, battery and storage constraints must be evaluated using the same run configuration. The absence of a brownout in a simulation is evidence about that model only.

A.5.5 Science Data, Storage and Downlink

The draft describes 64 histogram bins of 4 bytes each, yielding 256 bytes per histogram before packet headers, time tags, location, housekeeping or retransmission overhead. At one histogram per minute the bin payload alone is $256 \times 8/60 \simeq 34.1$ bit/s. The nominal daily data estimate in the draft is 300–400 kB/day, whereas the Science simulator's actual packet flush and mode availability depend on its separate output configuration. These values need a common packet definition before declaring storage or downlink closure. The Science default downlink rate is 19.2 kbps while the draft quotes 15–18 kbps. The maximum uncorrected raw contact capacity follows $D = R_{\text{link}} \sum_k t_{\text{contact},k}$, but usable data return must account for framing, packet losses, mission overhead, and science-data scheduling. The final RF link budget is addressed in Appendix A.6.

A.5.6 Cost and Procurement Budget

The initial MCR records a preliminary component cost table with most entries marked TBD. No consolidated priced procurement package is established in the supplied materials. The final bill of materials will require vendor quotations, quantities, testing and integration costs, contingency, delivery lead times and ownership of custom payload manufacturing. Until these inputs are supplied the cost budget remains preliminary.

A.6 Orbit, Coverage and Communications Supporting Analysis

A.6.1 Orbit Propagation and Orbital Lifetime

The mission baseline is a near-circular 600 km Sun-synchronous orbit at 97.6° inclination. The supporting simulation source contains several separate orbit models rather than a single uniform propagator. The annual `power/ANTARA_MCR.py` campaign runs 365-day numerical propagation across RAAN $0^\circ, 30^\circ, \ldots, 180^\circ$ with two Monte Carlo realizations each, 300-second reporting steps, 4.0 kg spacecraft mass and 97.4° inclination. Its force model includes gravity, DTM2000 atmospheric drag, solar radiation pressure and third-body perturbations. Its SAA and ground-station encounter classifiers are simplified geographic boxes. The dedicated `orbit/ANTARA_drag_finalconfig.py` assumes a 4.0 kg spacecraft, $C_D = 2.2$, projected area 0.076799 m^2 multiplied by a 1.5 tumbling factor, 98° inclination, and the January 2000 space-weather reference. It uses DTM2000, an RK4 step of 60 seconds and samples altitude daily. It computes duration from 500–600 km initial altitudes until crossing a *450 km study cutoff*; this is not an end-to-end calculation of atmospheric reentry or post-mission disposal compliance. The newer two-run-per-RAAN drag data to a nominal 300 km cutoff are separately

Table 23: Principal standalone orbital-study differences.

Study	Key configuration	Principal limitation
Annual RAAN campaign	365 days, 7 RAAN cases, 2 runs each; 4 kg, 97.4°	Simplified SAA/GS regions; run assumptions differ from DITL
Final-configuration decay	500–600 km, DTM2000, 4 kg, 450 km cutoff	No propagation from cutoff to physical reentry
Analytical CONOPS	30 days, Eckstein–Hechler, 60 s step	Rectangular station region, not actual elevation visibility
Pointing budget	One-day Keplerian orbit, 97.4°, RAAN 0°	+Z antenna body axis, unlike nominal +Y relay

reported in Appendix H. They do not retroactively change the definitions or parameters of the earlier 450 km study. A full disposal assessment must use the applicable mission and regulatory requirements, a defined physical end criterion, atmosphere/solar-activity uncertainty, and the final deployed spacecraft configuration. One-year altitude remaining above 550 km does not by itself demonstrate compliance.

A.6.2 SAA Boundaries and Science Coverage

The fast CONOPS and annual campaign use the simplified SAA geographic region $-50^\circ \leq$ latitude $\leq 0^\circ$ and $-90^\circ \leq$ longitude $\leq 40^\circ$. This differs from the science mapping region stated elsewhere in the report. A region crossing 0° longitude must have an unambiguous longitude convention; the stated box in the success criteria should be reconciled before computing cell counts or coverage percentages. The 30-day `conops/ANTARA_conops.py` tool uses an Eckstein–Hechler J2-perturbed orbit, 60-second samples and a rectangular Goa region defined by latitude 0–20° and longitude 60–80°. It logs eclipse and orbit classifications but does not compute true topocentric elevation. A SAA-box event, an SAA crossing, a unique orbit and an operational detector pass are different quantities and must not be described by the same count. In particular, the legacy annual result of over 8,000 “SAA-only orbits” cannot represent distinct revolutions of a 600 km spacecraft in a year; its classifier output requires interpretation against the underlying raw counters. The integrated Science simulation instead

maintains contact/pass histories and counts accepted detector events and science flushes. Its present proton-flux baseline is not a live AP9 solution; any predicted science precision must retain that limitation.

A.6.3 Ground Station Visibility and Pointing

The primary Goa station is approximately at 15°N, 74°E. The integrated Science default uses 70 m station altitude and a 15° elevation mask. The separate `pointing/ANTARA_pointing_budget.py` uses 30 m altitude and a 10° elevation mask for a one-day Keplerian case. Its antenna boresight is +Z in a nadir-pointing attitude; Science defines the relay antenna on +Y and slews to a moving ground-station line of sight. Results from the former must not be used to claim validated +Y relay acquisition in the latter. The Science code supports a second station when configured. Among simultaneously visible stations it selects the greatest elevation above each station's own mask, with primary-site tie-breaking. This is a modeled selection policy, not evidence that a second station has been selected or constructed.

A.6.4 Communications Link and Verification

The draft specifies a GomSpace NanoCom AX100 radio, a directional antenna and an omnidirectional backup concept; the final operating frequency plan, received sensitivity, modulation, antenna pattern, pointing loss, cable loss, polarization and full transmitter-to-ground receiver budget are not verified in the reviewed simulation source. The preliminary reported rates are 15–18 kbps in the draft and 19.2 kbps in the integrated model. Neither should be converted to confirmed daily delivered data volume without link and packet overhead analysis.

$$P_{\mathrm{rx,dBm}} = P_{\mathrm{tx,dBm}} + G_{\mathrm{tx,dBi}} + G_{\mathrm{rx,dBi}} - L_{\mathrm{path,dB}} - L_{\mathrm{other,dB}}. \tag{8}$$

The final system verification should include antenna impedance matching, S-parameter characterization, antenna gain/pattern or accepted equivalent evidence, RF-chain gain and noise figure, end-to-end receive testing at the intended frequency, and downlink throughput under realistic contact conditions. No numerical link margin is claimed without the corresponding hardware and test inputs.

Historical source-detail warning. Appendices B–E below preserve transcribed earlier material for traceability. Where they conflict with the revised main sections and Appendix A (especially the FPDU stack), the newer baseline controls; historical figures are not current flight selections.

B Source Orbit and Mission Timeline Detail

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

B.1 Mission Design Baseline and Assumption Register

Parameter	Baseline value	Status / source	Open reconciliation
Spacecraft wet mass	TBD	Preliminary	Check CAD and component budget
Battery energy capacity	TBD	Preliminary	Reconcile EPS product and DITL
Orbit epoch / RAAN	TBD	Trade-dependent	Specify scenario and epoch
Ground-station elevation mask	TBD	Preliminary	Match contact simulations

B.2 Mission-Level Design Trades and Decision Log

Decision	Options evaluated	Selection rationale/evidence	Status
Orbit / RAAN	TBD	Refer to Sec. 4	Open
Solar array / EPS	TBD	TBD	Open
Attitude actuation	TBD	TBD	Open
Communication architecture	TBD	TBD	Open

B.3 Development Milestones and Review Gates

Gate	Target date / status	Exit evidence and dependencies
PDR	TBD	TBD
CDR	TBD	TBD
Environmental qualification	TBD	TBD
Flight readiness / launch integration	TBD	TBD

C Source Subsystem Detail and Tables

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

The ANTARA spacecraft architecture has been developed as a modular, scalable, and mission-oriented CubeSat platform optimised for scientific radiation measurements withi n the South Atlantic Anomaly. The spacecraft follows a centralised avionics architecture in which the On-Board Computer (OBC) functions as the primary command. The communications subsystem, Payload Subsystem and ADCS subsystem work on the basis of the instantaneous mode. Following is the spacecraft system architecture.

C.1 System Architecture Overview

The ANTARA spacecraft is a **3U CubeSat**, conforming to the standard 3U form factor of 100 × 100 × 340.5 mm as defined by the CubeSat Design Specification (CDS) within the standard 3U mass allowance. Compliance with the applicable CDS revision, including deployer-imposed mass, center-of-gravity, and rail/standoff constraints, will be formally verified once a launch integrator and deployer are selected.

The five primary spacecraft subsystems integrated around a centralised avionics architecture are: On-Board Computer (OBC), Electrical Power System (EPS), Attitude Determination and Control System (ADCS), Communications System (COMMS), and Scientific Payload.

C.1.1 Electrical Power System (EPS)

The Electrical Power Subsystem is responsible for generating, storing, and distributing electrical power to all spacecraft loads. The COTS model we have taken for the simulation is the ISISPACE ICEPS2 Type A compact EPS, which includes an integrated 2-cell battery pack, autonomous battery heating, I2C communication, up to 6 output voltage domains, and protection features including overcurrent and reverse-current protection.

Table 27: ICEPS2 Type A parameters

Parameter	Preliminary Requirement
Mass	184 ± 5 g
Volume	96 × 92 × 26.45 mm
Energy storage	22.5 Wh (Type A specification)
Operating temperature	−20°C to +70°C
Interfaces	I2C housekeeping/control
Voltage domains	Regulated 3.3 V, 5 V, unregulated bus; up to 6 output domains

Note: The mission simulation uses a 25 Wh battery model as a conservative planning estimate. The final power budget must be reconciled against the ICEPS2 Type A 22.5 Wh specification at PDR.

Mode Power Estimates:

Table 28: Spacecraft operating-mode power estimates

Mode	Active Components	Estimated Power (W)
Cold & Dark Boot	EPS + OBC	0.8
Detumbling	EPS + OBC + ADCS MTQs	3–6
Safe Mode	EPS + OBC + Radio RX	1–2
Sun-Pointing	EPS + OBC + ADCS	2–3
Detection Mode	EPS + OBC + Payload + ADCS	6–9
Relay Mode	EPS + OBC + Radio TX + ADCS	4–7

Table 29: Preliminary subsystem electrical power budget

Subsystem	Idle / Standby (W)	Nominal (W)	Peak (W)
EPS	-0.093	0	0
Battery	-	charging capacity(28.8Wh)	-
OBC	-	-0.17	-0.9
Radio	-0.15 (rx)	-2.64 (tx)	-3 (TX max)
Unidirectional antenna	-0.04	-0.06	-2 (Deployment)
ADCS	-0.8	-2.6	-4.5
Payload (Assumed)	-	-3.0	-5.0
Solar Panels	-	11.5W	22W
GNSS Navigation	-0.21W	-0.255W	-0.39W

*Payload data is yet to be obtained, Worse case scenarios for each state is taken

C.1.2 Communications System (COMMS)

The Communications System provides the spacecraft's interface with the ground segment and supports telemetry transmission etc. The COTS model we have taken for the simulation is the GomSpace NanoCom AX100 radio system working on two antenna systems: GomSpace ANT430 omnidirectional antenna and ISIS UHF dipole antenna.

The primary ground station is planned for **Goa, India**.

Table 30: Goa ground station assumptions

Parameter	Value
Latitude	15°N
Longitude	74°E
Altitude	70 m
Elevation mask	15°

Simulation predicts an average of **two contacts per day**, with durations of approximately **368 s** and **414 s**. The maximum elevations are approximately **38.5°** and **77.5°**, providing roughly **780 s of total daily contact time**.

Although our simulations currently have one ground station, we aim to have more ground stations for receiving purposes.

Communications Concept

ANTARA uses a **Relay Mode** architecture, with communications restricted to ground-station

contact periods. During Relay Mode, the spacecraft performs:

- Telemetry transmission
- Science-data downlink
- Command reception
- Operational synchronisation

Downlink Architecture

The baseline downlink rate is **15–18 kbps**.

The scientific payload does not transmit individual proton events. Instead, the onboard FPGA accumulates events into **64-bin spatial-energy histograms**. Each histogram contains:

64×4=256 bytes

At one update per minute, the science data rate is approximately:

(256×8)/60≈34 bps

This corresponds to an estimated **300–400 KB/day** of science data.

With two daily contacts totalling approximately **780 s**, a 15 kbps link provides:

15,000×780≈11.7 Mbit

Thus, the available capacity provides approximately a 4× margin over the expected science-data volume, allowing for housekeeping, operational logs, attitude data, protocol overhead, and retransmissions.

The **9.6 kbps** value used in earlier analysis is superseded by the architecture-driven **15–18 kbps baseline**.

C.1.3 Payload subsystem

Overview

The payload is a **Flux Particle Detector Unit (FPDU)**: a compact proton spectrometer using a silicon ΔE front detector followed by a 5-layer EJ-200 plastic scintillator stack, read out by SiPMs, with a rear veto layer. It measures trapped protons from **5–100 MeV** in **64 log-spaced energy bins** at Sun-synchronous orbit (~600 km)

Physical Stack — Dimensions Along Boresight

Table 31: FPDU telescope physical-stack dimensions

Component	Thickness	Notes
Aperture tube + structure	30 mm	Sets θ_max = 18.4°
Si wafer	1 mm	Active detector
Si mount + PCB	3 mm	Carrier board
L1 + SiPM + coupling	4 mm	Scintillator** + SiPM package
L2 + SiPM + coupling	6 mm	
L3 + SiPM + coupling	14 mm	
L4 + SiPM + coupling	24 mm	
L5 + SiPM + coupling	32 mm	
Rear veto + SiPM	4 mm	
Optical isolation gaps	~3 mm	Black wrap between layers
Readout electronics board	15 mm	FPGA, ADCs, shaping, bias
Structural margin	5 mm	
Total	**~141 mm**	Fits in 2U payload volume

****All these calculations are done assuming EJ200 plastic scintillator and is subject to change.**

Dead Time and Livetime

Table 32: FPDU dead-time and livetime assumptions

Parameter	Value
Dead time per event	**~250 ns (shaping + ADC readout)**
Peak total event rate (SAA core)	**~100 Hz**
Dead time fraction at peak flux	**12.5%**
Livetime fraction at peak flux	**87.5%**
Dead time at SAA periphery	**< 0.1% (negligible)**

C.1.4 Attitude Determination and Control System (ADCS)

Attitude Determination and Control System(ADCS) continuously processes sensor data to calculate the spacecraft's exact position relative to external reference points. The COTS model we have taken for the simulation is TensorTech ADCS - 10m - D. The model uses a comprehensive suite of components: one ADCS-Main Control Board (ADCS-MCB), four TensorFSS-15M Fine Sun Sensors, and one integrated Control Moment Gyroscope (CMG) model.

Table 33: TensorTech ADCS-10m-D preliminary characteristics

Characteristic	Value
Pointing Knowledge	0.3 degree (general) 3-100 arcseconds (with startracker)
Pointing Accuracy	0.4 degree
Power consumption	0.8W (idle) 2.6W (nominal) 4.5W (Max)
Mass	450g

Control Modes and Algorithms

Detumbling. Following deployment, the ADCS executes a two-stage control law: an initial B-dot magnetic detumble to reduce high angular rates acquired during deployment, followed by reaction-wheel damping to bring the spacecraft to a stable, low-rate condition suitable for Sun Pointing Mode entry.

Simulation results show detumble completing in approximately 2,460 seconds from initial body rates of [6.0, −5.5, 4.5] °/s. No hard detumble timeout is currently defined.

Sun Pointing. The ADCS orients the spacecraft +Z face toward the Sun to maximize solar-array power generation, using reaction wheels for fine pointing and magnetorquers for momentum dumping.

Detection Mode Pointing. During SAA traversal, the ADCS aligns the +X axis (payload boresight) with the spacecraft velocity vector to support directional proton flux measurement and East-West asymmetry analysis. Simulation results show a mean attitude error of approximately 2°, against a threshold of $\leq$5° and a goal of $\leq$2°.

Relay Mode Pointing. The ADCS points the +Y face (directional relay antenna) toward the ground station to optimize downlink signal strength during scheduled contacts.

Slewing. Mode transitions between Sun Pointing, Detection, and Relay require dedicated slew maneuvers. The 24-hour DITL simulation recorded 18 slew events with slew angles ranging from 5.4° to 175.8°; large-angle slews were identified as consuming non-negligible time and should be factored into mission scheduling and duty-cycle analysis.

Saturation Handling. Reaction wheel momentum saturation is managed through magnetorquer-based momentum unloading. If saturation persists beyond a defined hold timeout and pointing diverges, the FSM commands a transition to Safe Mode as a fault-protection response, this is the second-highest-priority transition in the mode state machine after critical battery conditions

Table 34: ADCS simulation performance summary

Metric	Requirement / Result
Detumble completion	~2,460 s from [6.0, −5.5, 4.5] °/s
Detection Mode pointing error	<2°
Slew events (24-hr sim)	18
Slew angle range	5.4°–175.8°
Momentum management	Magnetorquer unloading + Safe Mode fallback

Pointing Budget summary:

Ground station contact pointing budget analysis is being mentioned. The result is for a simulation taken to Propagate 1-day Keplerian orbit (analytical, no perturbations) with nadir-pointing attitude, computes off-boresight angle between +Z body axis (antenna boresight) and ground station LOS vector during passes above 10° elevation mask. Outputs pass statistics and off-boresight angle plots.

C.1.5 On-Board Computer (OBC)

The On-Board Computer is the central computational and operational-control unit of the ANTARA spacecraft. The COTS model we have taken for the simulation is GomSpace Nanomind A3200.

Purpose

The OBC serves as the spacecraft's primary operational-management system and continuously monitors spacecraft status while coordinating all major mission activities.

At 3.3V bus channel, The OBC can run on various clock speeds from 8 - 32MHz depending on the state of the CubeSat with an average power consumption of 0.17W

Functional Responsibilities

The ANTARA OBC performs mission sequencing, state monitoring, command execution, health monitoring, payload control, data-storage management, and communications scheduling. Payload control is closely integrated with orbit propagation, ADCS orientation, SAA traversal prediction, and spacecraft power availability, enabling synchronised science operations during South Atlantic Anomaly crossings.

Flight Software Philosophy

ANTARA uses a state-machine-based flight software architecture designed to maximise reliability, deterministic behaviour, operational predictability, fault containment, and verification simplicity. Each operational mode corresponds to a defined spacecraft state. State transitions occur only when predefined operational conditions are satisfied, such as battery thresholds, spacecraft attitude stability, payload readiness, and subsystem health. The state-machine architecture simplifies mission simulation, operational verification, fault diagnosis, and long-duration autonomous operations.

C.2 Spacecraft Coordinate System

A spacecraft-fixed body-frame coordinate system is used throughout ANTARA mission operations. The body-frame definition is designed around payload operational geometry, solar-panel placement, communications requirements, and CubeSat deployment constraints.

Table 35: Spacecraft body-frame configuration

Axis	Function
+X	RAM face / payload boresight — primary payload pointing direction during Detection Mode
+Y	Directional relay antenna face (nominally nadir-pointing in default hold; pointed to ground station during Relay Mode)
+Z	Fixed solar panel face (pointed toward Sun during Sun Pointing Mode)
-Y	Deployable solar panel packed face
-Z	Omnidirectional communications antenna face (Safe Mode / beacon transmission)
-X	PPOD deployment face

C.3 Ground Segment & Mission Operations

Ground Station Hardware

Primary ground station: Goa, India (15°N, 74°E, 70 m altitude, 15° elevation mask).

Hardware baseline:

- **Antenna:** UHF Yagi or crossed-Yagi on an az-el tracking mount, matched to the NanoCom AX100 downlink band
- **RF front-end:** transceiver/SDR compatible with AX100 modulation, LNA at the antenna feed to minimize cable-loss impact on receive sensitivity
- **Backup/contingency station:** none currently defined, remains an open item.

Secondary Ground Station: We are yet to determine secondary ground stations. An ideal ground station would be at a polar location to accommodate for the SSO orbit. A temporary simulation was taken for Svalbard (78°N, 15°E, 50 m altitude, 5° elevation mask).

Ground Station Software

- **Orbit propagation & tracking:** TLE-driven antenna pointing; TLEs refreshed at minimum daily from CelesTrak/Space-Track, more frequently in the days immediately following deployment when orbit uncertainty is highest
- **Pass prediction:** AOS/LOS scheduling consistent with the two-contacts-per-day, ~780 s total daily contact time
- **Command uplink interface:** command construction, queuing, and uplink receipt confirmation
- Documented shift handover: spacecraft state, open anomalies, and next-pass command queue passed between operators

D Source Integrated Simulation Detail

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

The Day-in-the-Life (DITL) Simulation represents one of the most important mission-analysis and systems-engineering activities within the ANTARA mission framework. The DITL simulation acts as an integrated operational verification environment used to evaluate whether the spacecraft can successfully achieve its scientific objectives while operating within the physical, operational, and resource constraints of a 3U CubeSat platform.

Unlike isolated subsystem-level analysis, the DITL framework evaluates the spacecraft as a complete operational system in which orbit dynamics, spacecraft attitude, power generation, battery charging and discharge, payload operations, ground-station communications, and mission-mode transitions all interact simultaneously.

The baseline DITL simulation duration for Mission Concept Review activities is approximately 12 hours. The team has additionally completed a year-long simulation using Orekit and Python-based subsystem models.

The DITL simulation tracks the following operational metrics:

- spacecraft position, velocity, latitude, longitude, altitude, beta angle, eclipse status
- Battery energy and SOC, solar array power generation, spacecraft electrical load consumption, net power balance
- Attitude error, pointing accuracy, reaction wheel momentum, slew events
- SAA detection opportunities, science data volume generated
- Ground-station visibility, contact timing and duration, and cumulative data-return volume

.

Executive takeaways

- Operationally stable. The spacecraft completes startup detumble, repeated science and relay activities, and returns to SUN_POINTING after each activity. SUN_POINTING occupies 87.5% of the run.
- Attitude control is accurate outside slews. Steady-state 95th-percentile pointing and attitude error are both 0.4 deg. The larger error peaks are associated with commanded reorientation events.
- Slew execution is well matched to the model. There are 18 slews totaling 18.8 minutes. Actual duration differs from the estimate by only 0.02 s on average, with a maximum difference of 0.31 s.

Radiation operations reduce science availability

The run includes six SAA passes totaling approximately 44.9 minutes.

Across the five proton passes

- Average science-active fraction is approximately 62.7%
- The lowest science-active fraction is approximately 37.9%
- Other passes generally remain between approximately 66% and 71%

This suggests that the radiation protection logic is active but does not completely disable science collection during all proton environment.

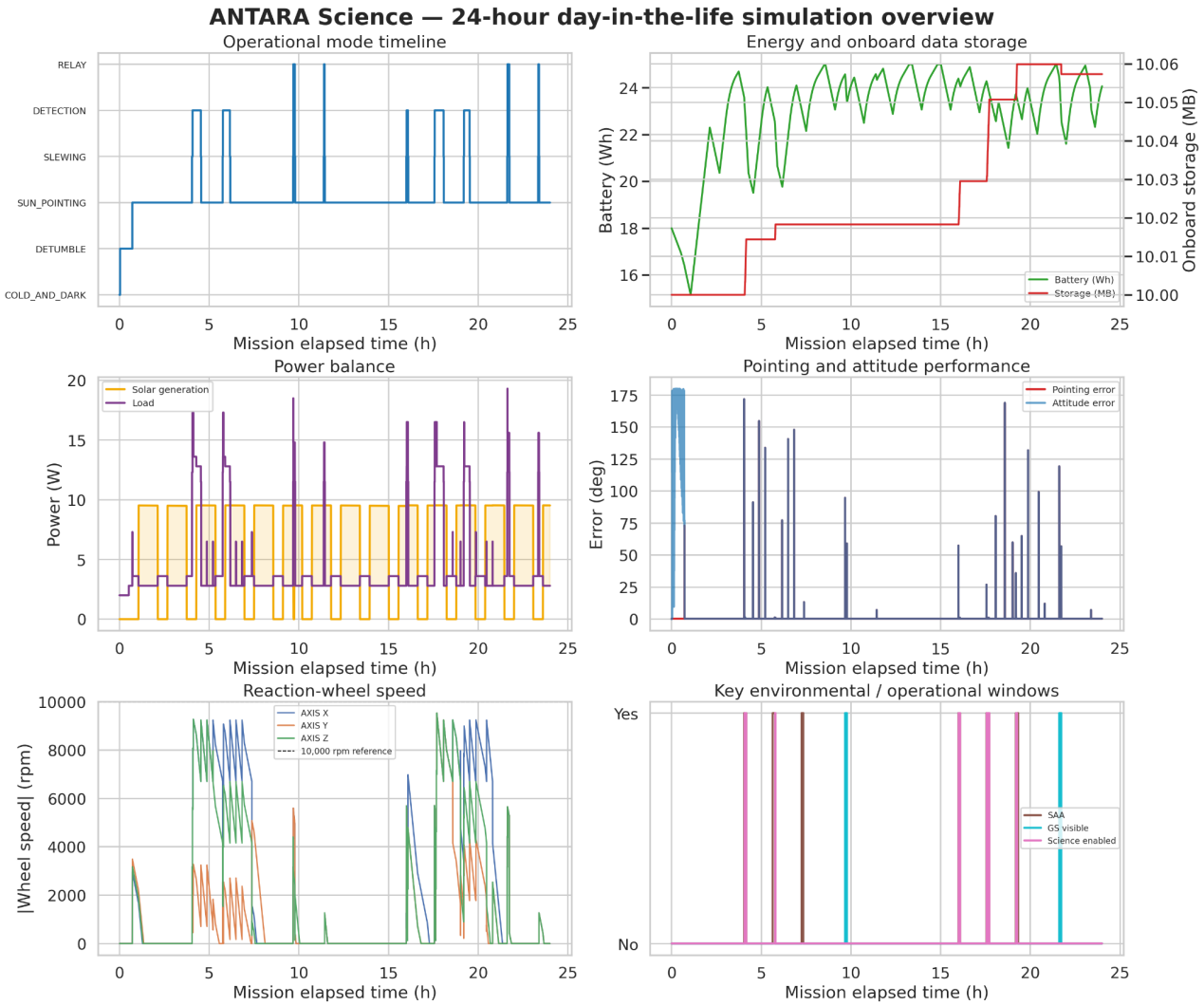


Figure 6: 24-hour integrated DITL overview: mode timeline, energy and storage, power, pointing, reaction-wheel speed and operational windows.

The battery never approaches depletion during this run.

- Initial battery energy is 18.00 Wh
- Minimum battery energy is 15.15 Wh
- Minimum level is approximately 84.2% of the initial value
- Final battery energy is 24.06 Wh
- Net battery gain over the simulation is approximately 6.06 Wh

Across the five proton passes

- Average science-active fraction is approximately 62.7%

- The lowest science-active fraction is approximately 37.9%
- Other passes generally remain between approximately 66% and 71%

Operational Mode Distribution (24 hours):

Table 36: 24-hour operational-mode distribution

Mode	Time Steps	Fraction of Day
Sun Pointing	37,781	87.5%
Detection	3,134	7.3%
Detumble	1,230	2.8%
Slewing	545	1.3%
Relay	450	1.0%
Cold & Dark	61	0.1%

Power and Energy:

- Average solar generation: 6.05 W
- Minimum battery energy recorded: 15.15 Wh (60.6% of 25 Wh planning capacity)
- No brownouts observed; battery remained above 30% safety threshold throughout the full 24-hour simulation.

SAA Operations:

- 6 SAA passes observed in 24 hours
- 5 of 6 passes achieved active payload science collection
- Total SAA dwell time: <2,700 seconds over 24 hours
- Science data generated: 0.060 MB (60 KB) per 24-hour period, consistent with 300–400 KB/day estimate at full nominal operations.

Ground Station Contact:

- 2 contacts with GS1_India (Goa, 15°N, 74°E) in the 24-hour window
- Contact 1: 368 seconds, maximum elevation 38.5°
- Contact 2: 414 seconds, maximum elevation 77.5°
- Daily downlink capacity at 15–18 kbps: ~1.5 MB/day — approximately 4× the expected daily science data volume.

ADCS Performance:

- Detumble completed in 2,460 seconds from initial body rates of [6.0, −5.5, 4.5] °/s
- 18 slew events recorded in 24 hours
- Slew angles ranged from 5.4° to 175.8°; large slews indicate some mode transitions require significant attitude manoeuvre time and should be factored into operational scheduling.

Proton Detection (5 active science passes):

- Mean measured FPDU: consistently 50–70% of AP9 model predictions across all energy channels (5–100 MeV), consistent with partial SAA coverage and pitch-angle sampling effects.
- Mean pitch angle ranged 15.6°–77.6° across passes
- Total counts per pass (Channel 0, lowest energy): 13,000–108,000 counts — well above all mission tier statistical thresholds.

One-Year Extended Simulation (RAAN 180° baseline):

Table 37: One-year integrated simulation statistics

Metric	Mean	Min	Max	99% CI
Average Power (W)	11.22	11.19	11.25	11.16–11.27
Average SOC	0.987	0.986	0.987	0.986–0.987
Minimum SOC	0.834	0.834	0.834	0.834–0.834
Battery Cycles	787	785	789	783–791

DITL Simulation Graphs

E Source Feasibility and Budget Detail

Historical source detail. Several numerical assumptions and design descriptions in this transcription have been superseded by Sections 1–5 and Appendix L. These tables are retained for traceability and are not the current approved mission baseline.

The following is retained from the original project draft for supporting detail and comparison. Numerical claims are preliminary unless specifically reconciled in the main text.

E.1 Integrated Engineering Budget Closure

Budget	Allocation / limit	Current total	Margin / evidence
Mass	TBD	TBD	TBD
Power (nominal / peak)	TBD	TBD	TBD
Battery energy	TBD	TBD	TBD
Science + housekeeping data/day	TBD	TBD	TBD
Downlink capacity/day	TBD	TBD	TBD
Volume / packaging	TBD	TBD	TBD
Cost	TBD	TBD	TBD

E.2 Verification and Validation Plan

Requirement ID	Verification type	Planned evidence / procedure	Phase / owner
TBD	Analysis / Test / Inspection / Demonstration	TBD	TBD

E.3 Technology Maturity and Development Gaps

Element	Current TRL / basis	Remaining maturation step	Target review
FPDU instrument	TBD	TBD	TBD
Integrated spacecraft	TBD	TBD	TBD
Ground RF chain	TBD	TBD	TBD

E.4 Open Issues, Owners and Closure Criteria

Issue ID	Decision / concern	Owner / deadline	Closure evidence
TBD	TBD	TBD	TBD

E.5 MCR Exit Assessment and PDR Entry Criteria

Concept feasibility established. **[TO COMPLETE:]** …

Remaining high-risk technical items. **[TO COMPLETE:]** …

Actions required before PDR. **[TO COMPLETE:]** …

F Integrated Mission Simulation and DITL Supporting Analysis

F.1 Purpose and Source

The integrated model in `src/simulations/ditl/ANTARA_Science_Sim.py` combines orbit propagation, spacecraft attitude, electrical power, mode transitions, proton-environment assumptions and detector accumulation, and ground-station communications. Its current implementation is a simulation model, not a hardware-in-the-loop qualification system. The shared configuration and individual presets are documented in Appendix A.3.

F.2 Defined Mission Scenarios

The source defines `leop_24h`, `no_leop_24h` and `leop_1yr`. The first includes Cold and Dark and Detumble; the second begins in Sun Pointing without LEOP startup; the annual scenario includes startup over 8,760 hours. Command-line values may override preset defaults. A comparison of the three scenarios is meaningful only when each output is accompanied by its exact version and configuration.

F.3 Mode Execution, Time and Energy

During startup, detumble and slewing, the simulator uses its fine time cadence. Other operating modes may use a coarser step. Mode occupancy, energy use, detector accumulation and mission elapsed time advance with the same actual integration interval. The interval length is recorded in `interval_duration_sec`; the last zero-duration terminal telemetry row should not be counted as additional mission time. Reported fractions should therefore use total elapsed time, not raw row fractions. The ADCS model retains finite slew and settling requirements, momentum saturation handling and recovery logic. During saturation hold, scientific collection and downlink transmission are inhibited in the documented model. The end of a planned slew duration is not by itself proof of achieved target alignment.

F.4 Science Data, Detector Statistics and Model Limits

The detector model accumulates expected bin counts and serializes histograms into science data separately. An interval with accepted events need not produce a positive new packet volume until a flush. SAA-exit and simulation-horizon flushes belong to the corresponding pass totals. Bin precision estimates use $1/\sqrt{N}$ for accumulated expected counts, but this does not represent calibrated detector response, stochastic-count realization or a fully validated low-count confidence interval. The environment model's legacy AP9-named variable is a configured baseline and not an independent AP9 computation.

F.5 Results and Acceptance Evidence

The source archive contains figures with scenario names but the final report must associate each approved graphic with its source revision, scenario configuration, complete CSV and summary. Earlier figures produced before corrected step accounting, attitude or wheel-momentum handling should not be mixed with current simulations without review. The code and documentation alone establish model capabilities and algorithms, not full-year completion, numerical accuracy or mission success.

Table 42: Integrated DITL results required for the final evidence package.

Scenario	Main quantities to extract	Evidence status
LEOP 24 hours	Start-up and detumble times, wheel behavior, minimum battery, first contact, SAFE events	Run-specific review required
Nominal 24 hours	Mode occupancy, SAA active time, packet accumulation, downlink, pointing error, power and battery extrema	Run-specific review required
Integrated one year	Year-end battery and orbit, total science and contact opportunities, cycle and anomaly statistics, run completion	Full-run verification required

G Additional MCR Study Evidence and Open Decisions

This appendix identifies the new data files supplied for the MCR. The plots and tabulated results are outputs of the supplied study files and do not establish final payload calibration, a new launch epoch, or compliance with spacecraft requirements by themselves.

G.1 Science grid and mission-tier studies

The MMSC, MSC and EMSC grid-disposition outputs are based on different cell grids and modeled geographic domains. “Primary Map” is the disposition assigned in each output file, not a declaration of global requirement closure. The table summarizes the file contents directly.

Table 43: Science-map cell dispositions in the supplied studies

Study	Evaluated	Primary Map	Supplementary	Unclassified
4-month MMSC	92	92	0	0
6-month MMSC	92	92	0	0
12-month MSC	2020	1915	105	0
14-month MSC	2020	1931	89	0
extended EMSC	7940	4213	3719	8

The MMSC grid file contains 92 modeled cells and the MSC files contain 2020 cells, while the EMSC file contains 7940 cells. These domains differ from the nominal whole-SAA bounding-grid cell totals in the mission success criteria. Coverage percentages must be recalculated against an agreed common region and approved cell eligibility rules before claiming satisfaction of a whole-region target.

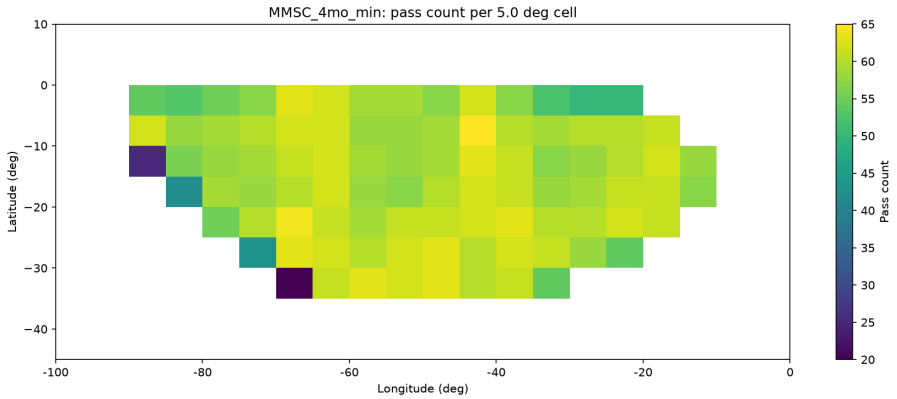


Figure 7: Four-month MMSC science-grid output from the supplied simulation archive; model-dependent.

G.2 Radiation and detector science model

The supplied IRENE-related inputs, proton-spectrum output, stopping-range plots and pile-up calculations support preliminary detector analysis. They do not establish a fully calibrated telescope geometry, dead-time limit, response matrix, or actual AP9 output from the integrated Science simulator. The integrated simulator’s AP9-named quantity is a configured baseline. Detector event-rate and scintillator dimensions remain open decisions.

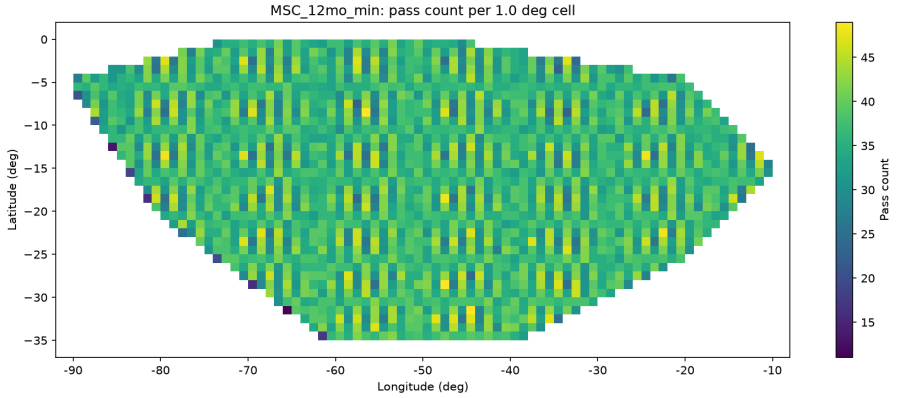


Figure 8: Twelve-month MSC science-grid output from the supplied simulation archive; model-dependent.

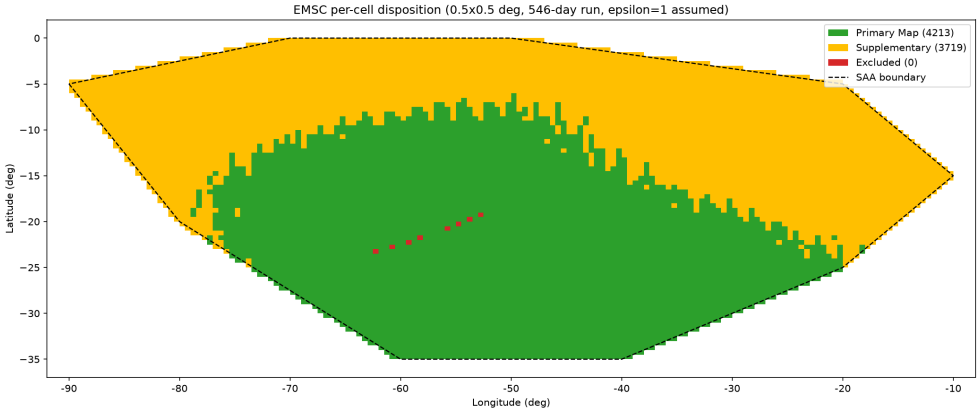


Figure 9: Extended science-grid classification from the supplied simulation archive; model-dependent.

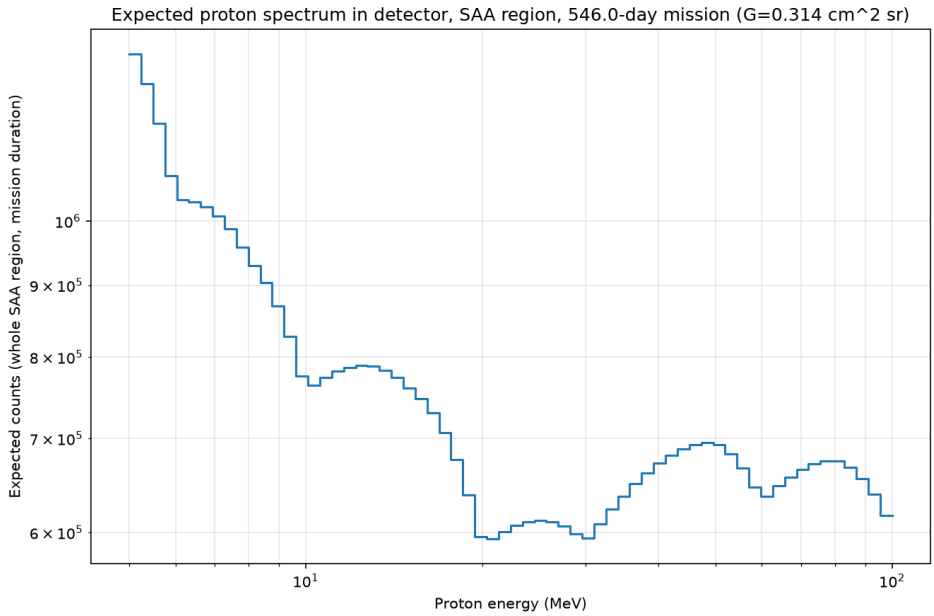


Figure 10: Supplied modeled SAA proton spectrum. This is a study output, not a validated detector measurement.

G.3 Ground-station and DITL source files

The supplied one-day run includes six recorded SAA passages, two Goa ground-station contacts, one initial detumble event lasting 2460 s, and eighteen slew events. Contact durations are approximately 368 and 414 s. This is a historical run with an epoch of March 2000; its performance is not automatically representative of a late-2028 or early-2029 mission epoch. The complete telemetry CSV and event tables are retained externally as raw evidence.

G.4 Preliminary procurement and mass allocation

The uploaded bill of materials identifies candidate ADCS, OBC, solar-array, EPS, structure, communications, navigation and antenna parts. The sum of currently specified component-mass entries is 1536.3 g. The payload and deployer-interface hardware do not have mass entries, and the deployable-panel quantity and overall packaging need review. Consequently the bill of materials cannot yet close the mission target of less than 3000 g. Procurement prices remain to be supplied.

G.5 Manufacturer documentation and CAD inputs

Manufacturer documents were supplied for the ISISPACE ICEPS2, TensorTech integrated ADCS, GomSpace NanoMind A3200 and NanoCom AX100, 2nd Space CORE/DROP/VERSE hardware and the proposed GNSS components. CAD inputs were supplied for several candidate components. Final mechanical envelopes, actuator configurations, harness allowances and interface drawings must be checked against the actual proposed configuration before the component baselines become final.

G.6 Remaining evidence placeholders

Table 44: Open evidence required for MCR closure

ID	Open item	Needed evidence
O-01	Reconcile historical SAA counter	New two-run orbit classifications are internally consistent; establish classifier geometry and event definitions
O-02	2028/2029 epoch and RAAN-independent assessment	Updated multi-RAAN simulation configurations
O-03	Long-term drag and disposal	Verify new 8–10-year cutoff runs, one truncated case, and post-cutoff reentry
O-04	Solar area, efficiency, degradation and EOL margin	Approved solar/EPS report from final configuration
O-05	FPDU event rate, energy channels and geometry	Payload geometry, detector-response and pile-up validation
O-06	Science-tier coverage requirements	Consistent geographic denominator, per-channel uncertainties and pass requirements
O-07	Ground station RF and data-rate closure	Final payload packets, coding, link and antenna design
O-08	Thermal and structural margins	Model outputs checked against component limits; tests planned
O-09	Complete spacecraft mass and cost	Payload, wiring, deployment items, quantities and quotations

ID	Open item	Needed evidence
O-10	Integrated long-term and scenario verification	Approved output files, code revision and run parameters

G.7 Bill of materials, as supplied

The following values are transcribed from the uploaded `BILL OF MATERIALS.xlsx`. Components are candidates, and none of the unresolved part numbers or interfaces is to be regarded as a final PDR selection. Values of "TBD" have not been imputed.

Table 45: Candidate component mass entries from supplied bill of materials

Subsystem	Component or part	Qty.	Mass (g)
ADCS	ADCS-10m	1	495
OBC	NanoMind A3200	1	24
Power Generation	Core 1Z	1	114
Power Generation	DROP	1	118
EPS	ICEPS 2S1P+iEPS Battery Type A	1	189
Structure	Verse-03	1	310
Communications	NanoCom AX100	1	24.5
Communications	UHF dipole antenna	1	90
Communications	ANT430	1	30
Deployment	TBD	1	TBD
Payload	Custom	1	TBD
Navigation	ZED-F9P-15B	1	1.8
Navigation	AHP54510.07.0100C	1	140

The workbook reports a total of 1536.3 g of currently entered masses, before adding the unallocated payload and deployment-interface mass. The current solar-panel quantity entries require confirmation against the intended one fixed and two deployable panel geometry. Costs in the workbook are not quoted amounts.

G.8 Datasheet alignment checks

The supplied ICEPS2 datasheet describes a 2S1P battery pack with nominal energy 28.8 Wh. The 25 Wh value remains the mission simulation planning capacity. The supplied TensorTech ADCS-10m datasheet explicitly describes a control-moment gyroscope configuration. Since ANTARA intends to use a reaction-wheel and magnetorquer integrated system, that datasheet cannot be treated as proof of the selected ADCS hardware configuration. A matching vendor model and specification must be supplied. The GNSS, radio, antenna, deployment-panel and structure datasheets are candidate mechanical/electrical references, pending interface selection.

G.9 October 2026 reporting baseline

The approved reference analysis epoch is 1 January 2029, and the satellite design maximum is 3.0 kg. Latest supplied simulator results are provisional until rerun with this epoch. The October FPDU Rev. 2.1 description used a single main scintillator block. The 4 November user geometry update specifies five main blocks plus a front scintillator; assembly and optical segmentation remain unresolved (Appendix A.2). The historical five-layer stopping response remains superseded. No SPENVIS/GRAS flight-orbit response output is yet represented as

completed. The science-operations objective remains about five years; the new drag output suggests roughly 8–10 years to its modeled cutoff, subject to configuration and physical-model verification (Appendix H).

H Updated RAAN Trade and Orbital Decay Evidence

This appendix transcribes the four latest supplied MCR campaign CSV files: `raan_master_trade_matrix(1).csv`, `mission_statistics(1).csv`, `drag_lifetime_analysis(1).csv`, and `orbit_classification(1).csv`. These outputs are the newest *provisional* trade-study evidence, not an integrated DITL rerun or a flight-verified performance guarantee. The approved mission analysis epoch is 1 January 2029; the supplied CSV files alone do not prove that they were generated with that epoch or with the final 3 kg spacecraft model.

H.1 RAAN trade-study summary

Table 46: Latest RAAN trade matrix, as provided. SOC is expressed as a percentage of capacity; orbit counts are sums across the study runs, not annual values.

RAAN	Power (W)	Mean SOC (%)	Min. SOC (%)	Life (yr)	SAA-only	GS-only
0°	11.02	97.9	69.6	9.46	79,377	11,536
30°	10.82	97.2	74.7	9.53	79,865	11,613
60°	10.28	95.1	70.4	8.95	74,917	10,995
90°	10.13	94.9	72.7	8.88	74,340	10,969
120°	9.02	88.1	54.4	8.23	69,267	10,196
150°	9.50	92.7	71.6	8.66	72,819	10,657
180°	11.17	98.2	73.4	8.58	72,103	10,498

The sampled RAAN set is $0°, 30°, 60°, 90°, 120°, 150°, 180°$. The largest tabulated mean generated power is 11.17 W at $180°$; the smallest is 9.02 W at $120°$. The latter has the lowest tabulated minimum SOC (54.4%). Consequently the current study does *not* justify naming $0°$ or $30°$ universally optimal. The choice of RAAN is launch-dependent; a robust spacecraft must be assessed across its feasible orbital-insertion cases. The RAAN matrix does not by itself demonstrate the fully integrated power/attitude/communications performance of a final 2029 mission.

H.2 Orbit-classification bookkeeping

The raw classification table contains 685798 rows: seven RAAN values and two recorded runs per RAAN. The entries in the RAAN trade matrix match the summed counts of `QUIET`, `ONLY_GS`, `ONLY_SAA` and `MIXED` orbit classes in that source file. `orbit_number` is unique within each RAAN/run pair. Classifications are mutually exclusive for each orbit record, but the tabulated counts pool both runs and therefore must not be presented as a one-year orbit count. The modeled classifier and event geometry still require documentation before they can be used as physical detector-on SAA passes or real GS contacts.

H.3 Lifetime to the modeled 300 km cutoff

Across the 14 records, modeled termination times span 8.23–9.99 years. Thirteen rows terminate near 300 km. The remaining $30°$ record terminates at 334.93 km after approximately 9.99 years, so it cannot be counted as a completed decay-to-300-km result. This suggests an approximately 8–10-year decay timescale *under these study assumptions*, not proof of the earlier five-year orbital-decay estimate. The output includes no post-300-km propagation to an atmospheric reentry condition. Atmospheric model, ballistic coefficient, starting epoch, mass, attitude-dependent area, solar activity, termination cause and parameter uncertainty require source-code provenance review. These results are preliminary and do not establish

Table 47: Two drag-study records per RAAN. Time is to termination of the supplied integration; the listed endpoint is not atmospheric reentry.

RAAN	Recorded time (yr), range	Endpoint altitude (km), range
0°	8.96–9.96	299.9–299.9
30°	9.06–9.99	299.9–334.9
60°	8.87–9.04	300.0–300.0
90°	8.86–8.90	299.6–299.8
120°	8.23–8.24	300.0–300.0
150°	8.53–8.79	299.9–299.9
180°	8.33–8.83	299.9–299.9

disposal compliance.

H.4 Monte Carlo statistics: scope and interpretation

The provided `mission_statistics(1).csv` corresponds numerically to the 180° RAAN row for average power (11.17485 W), mean SOC (0.98234), minimum SOC (0.73446), cycle count (6565.43) and lifetime (8.58241 years). Its confidence intervals should not be interpreted as 99% population coverage without verification of the statistical method and independent sample count; the raw study includes only two runs per RAAN. The power and lifetime fields represent modeled quantities, not measured solar electrical output or a verified physical lifetime.

H.5 Design implication for the MCR

The operational science campaign targets a roughly five-year mission, while the newest drag study points toward a potentially longer residence time before reaching 300 km. These statements are not interchangeable. The MCR should maintain the five-year *operational objective* while qualifying the uncertain passive decay and post-mission disposal timeline. Detailed DITL operations are being rerun separately; their future results should be incorporated with independent identification of date, simulation revision and model assumptions.